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**EKF-BASED SENSOR FUSION FOR
MECANUM WHEEL ROBOT
LOCALIZATION USING ARUCO MARKER**

MAJOR: AUTOMATION CONTROL

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ABSTRACT

This project presents a 2D pose estimation method for a low-cost Mecanum-wheeled robot without relying on expensive Lidar or GPS systems. The purpose is to estimate the robot’s position and orientation by fusing angular velocity measurements from wheel encoders with data from an Inertial Measurement Unit (IMU). Due to the 45° -oriented rollers, Mecanum wheels inherently suffer from significant lateral slippage, leading to rapid odometry drift. To mitigate this issue, we implemented a multi-sensor fusion system using an ESP32-based hardware architecture running FreeRTOS. The core of the proposed solution is a 3-state Extended Kalman Filter (EKF) that integrates wheel encoders, a Z-axis gyroscope, and visual feedback from a USB camera.

The EKF state vector models the 2D pose (x, y, θ) . A rigorous statistical methodology was introduced to define the noise covariance matrices: the process noise matrix $\mathbf{Q}$ was determined through empirical slip tests, while the IMU measurement noise matrix $\mathbf{R}_{\text{IMU}}$ was characterized using first-difference variance analysis to isolate white noise from bias drift. For visual measurements, the covariance estimation involved measuring static noise ($\mathbf{R}_{\text{static}}$) at various distances and subsequently applying a dynamic scaling factor α , derived via the ordinary least squares method, to account for motion blur and mechanical vibrations.

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CHAPTER 1

INTRODUCTION

1.1 Motivation

1.1.1 Background and Practical Needs

In the context of the Industry 4.0 revolution, Autonomous Guide Vehicles (AGV) are being extensively deployed in fields such as logistics, automated manufacturing, smart warehousing, and internal goods delivery. The core requirement of a mobile robot system is the ability to autonomously determine its position within an environment and navigate to desired targets with high precision, even in the presence of environmental changes or cumulative sensor errors.

The Mecanum-wheeled robot is a particularly suitable choice for confined spaces due to its holonomic motion capabilities: it can move forward, backward, slide laterally, and rotate on the spot without needing to change its heading. This provides superior maneuverability compared to differential drive or Ackermann steering robots in complex indoor navigation scenarios.

However, Mecanum wheels also pose significant challenges in state estimation: the phenomenon of lateral slip causes pure odometry to accumulate errors rapidly, particularly along the Y-axis. Therefore, it is essential to integrate multiple sensor sources to achieve reliable position estimation.

The Extended Kalman Filter (EKF) is a prominent state estimation method that allows for the fusion of the robot's kinematic model with measurements from heterogeneous sensors such as encoders, IMUs, and cameras. Meanwhile, Aruco marker, a computer vision system based on 2D barcodes, provides highly accurate absolute positioning when the robot is in proximity to a marker. These tags serve as "anchors" to eliminate accumulated drift during long-distance travel phases.

This project develops a comprehensive localization and navigation system for a 4-wheel Mecanum robot, capable of performing multi-station missions in indoor environments. The architecture utilizes an ESP32 as the

embedded microcontroller and a PC running Python to handle the EKF, Aruco marker processing, and high-level control logic.

1.1.2 Limitations of Individual Sensory Methods

To compensate for the limitations of odometry, AGV systems typically fuse multiple complementary sensors. However, each sensor type has specific limitations when operating independently:

- **Encoders (Odometry):** Provide continuous high-frequency velocity information (100 Hz in this system) independent of environmental conditions. However, errors accumulate over distance and time due to wheel slip.
- **Inertial Measurement Unit (IMU/Gyroscope):** Measures instantaneous angular velocity with good short-term accuracy, complementing odometry in heading estimation. However, gyroscope bias, even after calibration, still accumulates over time.
- **Camera for Aruco Marker Recognition:** Provides absolute position information with an error of less than 1 cm at a distance of 0.5 m. This is the most valuable source of information regarding accuracy. However, the camera can only observe the marker when it falls within its Field of View (FOV).

1.1.3 Problem Statement

Based on the above analysis, the specific problem of this thesis is formulated as follows:

Precise, real-time localization is a fundamental prerequisite for autonomous mobile robot navigation in indoor environments where GPS signals are unavailable. For Mecanum-wheeled platforms, full holonomic mobility-enabling lateral translation and zero-radius rotation without steering wheel angles-comes at the trade-off of significantly higher wheel slippage compared to conventional differential-drive robots. Consequently, localization relying solely on wheel odometry (dead reckoning) accumulates estimation errors rapidly and unboundedly over time.

Visual landmarks (Aruco markers) mitigate cumulative drift by providing absolute pose measurements independent of motion history; however, they operate at lower update rates and experience intermittent outages during occlusions or when markers exit the camera's field of view. The core

problem addressed in this work is to fuse Mecanum wheel odometry, Inertial Measurement Unit (IMU) data, and ArUco visual observations into a unified localization system that maintains a high update frequency while effectively bounding cumulative estimation drift-simultaneously ensuring that the estimator faithfully reflects its true operational confidence, yielding pose estimates that are not only accurate but also statistically consistent.

1.2 Related Work

- **Kou and Haggemiller (2023)** in their work, "Extended Kalman Filter State Estimation for Autonomous Competition Robots," presented the application of an Extended Kalman Filter (EKF) to fuse data from odometry, physical models, and computer vision for Mecanum-wheeled robots. They demonstrated that while odometry provides high-frequency velocity data, it suffers from cumulative errors; conversely, computer vision provides accurate but intermittent global positioning. Their simulation results showed that integrating odometry with physical models significantly reduces the Root Mean Square Error (RMSE) of velocity. This study serves as a vital reference for configuring the process noise covariance matrix $\mathbf{Q}$ and measurement noise covariance matrix $\mathbf{R}$ through systematic tuning.
- **Faralli, A. et al. (2016):** "Indoor real-time localisation for multiple autonomous vehicles fusing vision, odometry and IMU data." This paper proposes a real-time localization system for multi-robot formations based on fusing data from fixed cameras, odometry, and IMU through two cascaded EKF filters. The first filter processes high-frequency odometry/IMU data, while the second corrects the pose when visual data becomes available. This fusion reduced positioning error by 70% (from 51 cm to 10 cm). The method also addresses visual system latency (300 ms) by utilizing a buffer to store historical data.
- **Yoon, S. W. et al. (2015):** "Kalman Filter Sensor Fusion for Mecanum Wheeled Automated Guided Vehicle Localization." This research focuses on addressing the wheel-slip problem inherent in Mecanum-wheeled AGVs. The authors employ a Kalman filter to fuse data from encoders and the StarGazer image sensor. Experimental results demonstrate that the system effectively eliminates cumulative encoder errors and anomalous landmark identification. This provides a foundation for understanding

the Jacobian kinematic model of Mecanum robots when handling slip phenomena.

- **Faria, S. et al. (2020):** "Sensor Fusion for Mobile Robot Localization using Extended Kalman Filter, UWB ToF and ArUco Markers." This study evaluates the effectiveness of an EKF in fusing odometry with Ultra-WideBand (UWB) distance measurements and ArUco angular data. By utilizing Mahalanobis distance to reject outliers, the system achieved a positioning accuracy of approximately 1.98 cm, outperforming standard indoor GPS or Wi-Fi solutions.
- **Kam, H. C. et al. (2018):** "An Improvement on ArUco Marker for Pose Tracking Using Kalman Filter." This document presents a linear Kalman filter to post-process ArUco pose data to reduce noise and jitter. The filter employs a constant velocity model to estimate the robot's pose during temporary occlusions. The study emphasizes the use of Quaternions instead of Euler angles in the EKF to avoid the "gimbal lock" phenomenon.
- **Joon, A. and Kowalczyk, W. (2023):** "Leader-Follower Approach for Non-Holonomic Mobile Robots Based on Extended Kalman Filter Sensor Data Fusion." The authors propose a leader-follower control model using an Intel RealSense camera on a rotating base. Utilizing Behavior Trees, the system switches between observing leader-mounted ArUco tags and fixed environment landmarks. The fusion of camera, encoder, and IMU data via EKF expands the robot's perception range without requiring additional hardware.

1.3 Scope

This project aims to achieve specific, quantitatively verifiable objectives:

- **Objective 1:** Develop real-time control firmware on the ESP32 microcontroller using a dual-core FreeRTOS architecture at a 100 Hz control frequency, ensuring stable reading of encoders and the IMU, and executing a four-wheel velocity PI controller with feedforward.
- **Objective 2:** Implement three-state Extended Kalman Filter (EKF) $[x, y, \theta]$ fusing three asynchronous sensor sources, with the Q and R matrices determined through methodological experiments, and verify consistency using the Normalized Innovation Squared (NIS) test.

- **Objective 3:** Camera calibration, marker detection and design a layout system of three Aruco markers to establish a unified world coordinate system.
- **Objective 4:** Deploy EKF position estimation on mecanum wheeled robot, testing and evaluating filter results.

1.4 Thesis Structure

This thesis is organized into six logically sequential chapters.

Chapter 1 – Introduction (the current chapter) presents the research background, analyzes the technical challenges of the multi-destination Mecanum robot localization problem, reviews related literature, identifies the research gap, and establishes the objectives and scope of the project.

Chapter 2 – Theoretical Background provides the mathematical foundation necessary for the entire system: Mecanum robot kinematics (forward and inverse kinematics, odometry), three-state EKF theory (process model, Jacobians, predict-update cycle, Joseph form), methods for determining the Q matrix from random walk and R_{imu} from first difference variance, NIS test theory based on the χ^2 distribution, Aruco recognition principles and coordinate frame transformations. **Chapter 3 – Hardware Design** describes the system’s two-layer distributed architecture, detailing the hardware components (Mecanum-wheeled robot, ESP32, L298N motor driver, IMU MPU-6050, USB-camera), and specifically the layout of three Aruco markers that establish the world coordinate system for the entire workspace.

Chapter 4 – Software Design details the implemented algorithms: ESP32 firmware with a dual-core FreeRTOS architecture, PI + feedforward controller, procedures for determining and storing Q/R matrices, Python EKF implementation with three update methods, coordinate frame converter.

Chapter 5 – Experiments and Evaluation Chapter 5 - Experiments and Evaluation validates the EKF fusion framework through two closed-loop reference trajectories, a square path and a circular path, each executed under Predict-Only (dead-reckoning) and Fusion mode.

Chapter 6 – Conclusion and Future Work summarizes the main contributions of the project, evaluates the advantages and disadvantages of the developed system, and proposes future work to improve and optimize the outcome of the system.

CHAPTER 2

THEORETICAL BACKGROUND

2.1 Mecanum Robot Kinematics

2.1.1 Structure and Principles of Mecanum Wheels

The Mecanum wheel was invented by Swedish engineer Bengt Ilon in 1972 and patented in the United States in 1975. The basic structure consists of a rigid wheel rim with a series of small rollers attached around its circumference, with their axes of rotation angled at 45° relative to the main axis of the wheel. When the wheel rotates, the contact point between the roller and the floor generates a friction force at a 45° angle neither purely in the forward direction like conventional wheels nor purely in the lateral direction.

By configuring four wheels with symmetrical tilt angles-the Front-Left (FL) and Rear-Right (RR) wheels having rollers angled in one direction, while the Front-Right (FR) and Rear-Left (RL) wheels are angled in the opposite direction-the robot achieves full 3-DOF holonomic mobility in a plane: translation along the X-axis (forward/backward), translation along the Y-axis (left/right), and rotation around the Z-axis in place. This is a capability that differential drive robots lack, as they must rotate 90° before moving laterally.

Lateral slip characteristics and impact on localization: The primary disadvantage of Mecanum wheels compared to conventional wheels is their high lateral slip coefficient. When the robot moves laterally ($v_y \neq 0$), the rollers do not roll perfectly on the floor but exhibit a sliding component. Taheri et al. [8] quantitatively measured that on standard flat floors, the lateral slip coefficient is 2 to 4 times higher than the longitudinal slip coefficient. This means that over the same duration, the encoders measure fewer pulses than the actual distance traveled when the robot moves laterally. A direct consequence is that the measurement noise matrix R_{enc} of the EKF requires $R[v_y, v_y] \gg R[v_x, v_x]$ to accurately reflect this characteristic, a crucial distinction from standard EKF models for differential robots.

2.1.2 Coordinate System and Sign Conventions

In this project, the robot frame coordinate system is defined as follows: the $+X$ axis points toward the front of the robot (forward direction), the $+Y$ axis points to the left of the robot, and the $+Z$ axis points upward (following the right-hand rule). Based on this, the sign conventions for velocities are established and experimentally verified:

- $v_x > 0$: robot moves forward.
- $v_x < 0$: robot moves backward.
- $v_y > 0$: robot moves to the left (along $+Y$).
- $v_y < 0$: robot moves to the right.
- $\omega_z > 0$: robot rotates counter-clockwise (CCW) - viewed from above.
- $\omega_z < 0$: robot rotates clockwise (CW).

The wheel ordering throughout the entire system is $[FL = 0, FR = 1, RR = 2, RL = 3]$, following a sequential count starting from the front-left wheel in a clockwise direction when viewed from above.

2.1.3 Inverse Kinematics

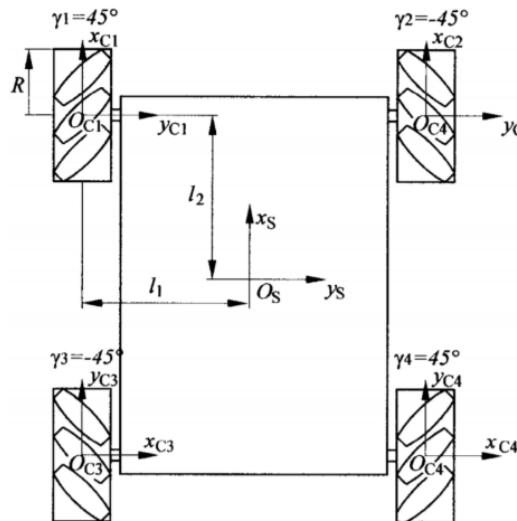


Figure 2.1: Mecanum kinematic

Inverse kinematics solves the following problem: given a desired robot velocity vector $[v_x, v_y, \omega_z]$ in the robot frame, calculate the required angular

velocity for each wheel. By analyzing the force vectors at the contact point of each wheel and ensuring the total force and moment equal the desired force, the inverse kinematics matrix is derived as follows:

$$\begin{aligned}
\omega_{FL} &= \frac{1}{R} [+v_x - v_y - (l_x + l_y)\omega_z] \cdot \frac{60}{2\pi} \quad [\text{RPM}] \\
\omega_{FR} &= \frac{1}{R} [+v_x + v_y + (l_x + l_y)\omega_z] \cdot \frac{60}{2\pi} \quad [\text{RPM}] \\
\omega_{RL} &= \frac{1}{R} [+v_x - v_y + (l_x + l_y)\omega_z] \cdot \frac{60}{2\pi} \quad [\text{RPM}] \\
\omega_{RR} &= \frac{1}{R} [+v_x + v_y - (l_x + l_y)\omega_z] \cdot \frac{60}{2\pi} \quad [\text{RPM}]
\end{aligned} \quad (2.1)$$

Given the robot's geometric parameters in this project: $R = 0.0335$ m (wheel radius), $l_x = 0.1075$ m (half front-back wheelbase), $l_y = 0.1025$ m (half left-right wheelbase), and $(l_x + l_y) = 0.2100$ m.

2.1.4 Forward Kinematics

Forward kinematics solves the inverse problem: from the measured angular velocities of the four wheels via encoders, calculate the robot's actual velocity. The encoders measure Δticks in each period dt , from which the wheel angular velocity (rad/s) is computed:

$$\omega_i = \frac{2\pi \cdot \Delta\text{ticks}_i}{\text{CPR} \cdot dt} \quad (2.3)$$

With $\text{CPR} = 4,532$ pulses/revolution ($= 11 \text{ PPR} \times 4 \text{ quadrature} \times 103 \text{ gear ratio}$). The robot velocity is calculated using the pseudo-inverse of the inverse kinematics matrix, which in this special case yields a closed-form solution:

$$v_x = \frac{R}{4} (\omega_0 + \omega_1 + \omega_2 + \omega_3) \quad (2.4)$$

$$v_y = \frac{R}{4} (-\omega_0 + \omega_1 - \omega_2 + \omega_3) \quad (2.5)$$

$$\omega_z = \frac{R}{4(l_x + l_y)} (-\omega_0 + \omega_1 + \omega_2 - \omega_3) \quad (2.6)$$

2.1.5 Euler Integration and Odometry

From the robot's velocity $[v_x, v_y, \omega_z]$ in the robot frame, the pose $[x, y, \theta]$ in the world frame is updated via Euler integration:

$$\theta(k+1) = \theta(k) + \omega_z \cdot dt \quad (2.7)$$

$$x(k+1) = x(k) + [v_x \cos \theta(k) - v_y \sin \theta(k)] \cdot dt \quad (2.8)$$

$$y(k+1) = y(k) + [v_x \sin \theta(k) + v_y \cos \theta(k)] \cdot dt \quad (2.9)$$

Where dt is extracted from the `dt_ms` field in the ESP32 telemetry rather than a fixed value, ensuring precise integration even with jitter in the control loop. The rotation matrix $[\cos \theta, -\sin \theta; \sin \theta, \cos \theta]$ transforms velocities from the robot frame to the world frame, introducing non-linearity into the process model-this is exactly why an EKF is required instead of a linear Kalman Filter.

2.2 Aruco Marker Detection and Pose Estimation

2.2.1 Pinhole Camera Model

The camera is modeled using the pinhole model, which maps a 3D point $\mathbf{P} = (X, Y, Z)$ to a 2D image pixel $\mathbf{p} = (u, v)$:

$$\begin{aligned} u &= f_x \cdot \left(\frac{X}{Z} \right) + c_x \\ v &= f_y \cdot \left(\frac{Y}{Z} \right) + c_y \end{aligned} \quad (2.1)$$

$$K = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} \quad (2.2)$$

where f_x and f_y denote the focal lengths along the two image axes expressed in pixels, and (c_x, c_y) represents the principal point (image center coordinates). Because real-world optical lenses inherently exhibit a certain degree of geometric distortion, the full camera model further incorporates radial and tangential distortion coefficients. These intrinsic parameters are determined through a formal camera calibration procedure, most commonly implemented using Zhang's method based on multiple views of a planar checkerboard pattern. Beyond the ideal model, physical lenses introduce distortion, including radial and tangential distortion, parameterized by the coefficient vector $D = [k_1, k_2, p_1, p_2, k_3]$. Both K and D must be determined via a calibration process before they can be used for pose estimation.

2.2.2 Aruco Marker Structure

Aruco marker is a visual fiducial system designed for rapid detection and precise pose estimation. The structure of a tag comprises:

- **Outer white border:** Creates a buffer zone to isolate the tag from the surrounding background.
- **Inner black border:** Defines the exact physical boundary of the tag.
- **Data payload:** A binary bit grid encoding the unique tag ID.
- **Four corners:** Reference points used to compute the homography and subsequent pose estimation.

The tag detection pipeline for each image frame consists of the following steps: grayscale conversion $\rightarrow$ adaptive thresholding $\rightarrow$ quadrilateral detection and filtering (based on area and orthogonality) $\rightarrow$ tag region normalization to a standard size (via homography) $\rightarrow$ payload decoding $\rightarrow$ sub-pixel refinement of the four corner positions.

2.3 IMU MPU6050

The MPU6050 is a 6-degree-of-freedom (6-DOF) inertial measurement unit (IMU) that integrates a 3-axis accelerometer and a 3-axis gyroscope on a single chip, communicating with the microcontroller via the I²C protocol.

The accelerometer measures the specific force, which combines true linear acceleration and gravitational acceleration. The measurement model is expressed as:

$$a_{\text{meas}} = a_{\text{true}} + g + b_a + n_a \quad (2.3)$$

where b_a denotes the accelerometer bias exhibiting slow drift over time and temperature variations, and n_a represents additive Gaussian white noise.

The gyroscope measures the instantaneous angular velocity about three orthogonal axes, following an analogous measurement model:

$$\omega_{\text{meas}} = \omega_{\text{true}} + b_g + n_g \quad (2.4)$$

Due to the persistent presence and slow drift of the bias term b_g , direct numerical integration of ω_{meas} to compute orientation yields unbounded error accumulation (gyro drift) unless periodically corrected. Within the proposed EKF framework, the vertical angular velocity (z-axis) measured by the gyroscope serves as an auxiliary observation source in the heading angle (θ) prediction model, fused alongside the Mecanum wheel kinematic updates to mitigate complete reliance on wheel slip-induced estimation errors.

2.4 Theoretical Foundation of the Extended Kalman Filter (EKF)

The Extended Kalman Filter (EKF) is the standard implementation of the Kalman filter for non-linear systems. In autonomous robotics, kinematic equations frequently involve trigonometric functions; thus, the EKF is employed to estimate the state by linearizing the system around the current state estimate using Taylor expansions.

2.4.1 Non-linear System Model

A non-linear dynamic system in discrete time is described by two fundamental stochastic equations:

1. **Process Model (State Equation):**

$$\mathbf{x}_k = f(\mathbf{x}_{k-1}, \mathbf{u}_k) + \mathbf{w}_k \quad (2.5)$$

2. **Measurement Model (Observation Equation):**

$$\mathbf{z}_k = h(\mathbf{x}_k) + \mathbf{v}_k \quad (2.6)$$

Where:

- $\mathbf{x}_k$: State vector at time k (e.g., position, orientation, velocity).
- $\mathbf{u}_k$: Control input vector (e.g., commanded wheel velocities).
- $\mathbf{z}_k$: Observation vector from sensors (e.g., IMU, Aruco marker, Encoders).
- $\mathbf{w}_k, \mathbf{v}_k$: Process and measurement noise, assumed to be zero-mean Gaussian white noise with covariance matrices $\mathbf{Q}$ and $\mathbf{R}$, respectively.

2.4.2 Linearization via Jacobian Matrices

Because the functions f and h are non-linear, the EKF performs a first-order Taylor expansion at each time step. The resulting Jacobian matrices represent the local linear behavior of the system:

• **State Transition Jacobian (F):**

$$\mathbf{F}_k = \left. \frac{\partial f}{\partial \mathbf{x}} \right|_{\hat{\mathbf{x}}_{k-1}, \mathbf{u}_k} \quad (2.7)$$

• **Measurement Jacobian (H):**

$$\mathbf{H}_k = \left. \frac{\partial h}{\partial \mathbf{x}} \right|_{\hat{\mathbf{x}}_{k|k-1}} \quad (2.8)$$

2.4.3 The EKF Recursive Cycle

The EKF operates through a recursive two-step process: *Prediction* and *Correction*.

Phase 1: Prediction (Time Update)

The filter projects the current state and error covariance forward in time to obtain the *a priori* estimate for the next time step:

1. **State Prediction:**

$$\hat{\mathbf{x}}_{k|k-1} = f(\hat{\mathbf{x}}_{k-1}, \mathbf{u}_k) \quad (2.9)$$

2. **Covariance Prediction:**

$$\mathbf{P}_{k|k-1} = \mathbf{F}_k \mathbf{P}_{k-1} \mathbf{F}_k^T + \mathbf{Q} \quad (2.10)$$

As the robot moves without external measurement updates, the error covariance $\mathbf{P}$ increases, representing growing uncertainty.

Phase 2: Correction (Measurement Update)

When a sensor measurement (such as an Aruco marker detection) becomes available, the filter corrects the *a priori* estimate to produce the *a posteriori* state:

1. **Innovation (Measurement Residual):**

$$\mathbf{y}_k = \mathbf{z}_k - h(\hat{\mathbf{x}}_{k|k-1}) \quad (2.11)$$

2. **Innovation Covariance:**

$$\mathbf{S}_k = \mathbf{H}_k \mathbf{P}_{k|k-1} \mathbf{H}_k^T + \mathbf{R} \quad (2.12)$$

3. **Optimal Kalman Gain:**

$$\mathbf{K}_k = \mathbf{P}_{k|k-1} \mathbf{H}_k^T \mathbf{S}_k^{-1} \quad (2.13)$$

4. **State Update:**

$$\hat{\mathbf{x}}_k = \hat{\mathbf{x}}_{k|k-1} + \mathbf{K}_k \mathbf{y}_k \quad (2.14)$$

5. **Covariance Update:**

$$\mathbf{P}_k = (\mathbf{I} - \mathbf{K}_k \mathbf{H}_k) \mathbf{P}_{k|k-1} \quad (2.15)$$

2.4.4 Parameter Significance

The performance of the EKF is highly sensitive to the tuning of the noise matrices:

- **Matrix Q:** Higher values indicate lower confidence in the process model (e.g., significant wheel slip), causing the filter to rely more heavily on sensor measurements.
- **Matrix R:** Higher values indicate lower confidence in sensor data (e.g., noisy camera image), causing the filter to follow the predicted model more closely.

2.5 Covariance matrices Q and R

2.5.1 Definition and Mathematical Role of Matrix Q in EKF

The process noise covariance matrix Q_k models the uncertainty arising from system dynamic unmodeled phenomena and unmeasurable external disturbances. Within the Extended Kalman Filter (EKF) algorithm, Q_k explicitly enters during the covariance prediction step:

$$P_{k|k-1} = F_k P_{k-1|k-1} F_k^\top + Q_k \quad (2.16)$$

where:

- $P_{k|k-1}$ represents the predicted state error covariance matrix.
- F_k is the Jacobian matrix of the non-linear state transition function $f(x, u)$.
- Q_k compensates for the cumulative growth of state uncertainty over each time step due to modeling inaccuracies.

The influence of Q_k on the filtering response is regulated through the Kalman Gain matrix K_k :

$$K_k = P_{k|k-1} H_k^\top (H_k P_{k|k-1} H_k^\top + R_k)^{-1} \quad (2.17)$$

- **High Q_k assignment:** $P_{k|k-1}$ increases $\rightarrow K_k$ increases $\rightarrow$ the EKF reduces reliance on the prediction model (odometry) and relies more heavily on correction updates from exteroceptive/proprioceptive sensors (e.g., IMU, LiDAR, Visual SLAM).
- **Low Q_k assignment:** $P_{k|k-1}$ decreases $\rightarrow K_k$ decreases $\rightarrow$ the EKF places greater trust in the encoder-based kinematic prediction model, thereby smoothing out high-frequency measurement noise.

2.5.2 Practical Considerations in Mecanum Wheel Robot Odometry

For odometry estimation relying on Mecanum wheel encoders, the physical system inherently introduces complex process noise dynamics that must be properly accounted for within Q_k :

- **Wheel Slip Phenomena:** The 45° passive rollers on Mecanum wheels cause inherently unstable friction forces. Under slippage, encoders continue registering pulses while true displacement deviates, creating significant discrepancies between idealized kinematics and physical motion.
- **Mechanical Kinematic Imperfections:** Mechanical vibrations, dynamic variations in effective wheel radius r due to rubber compliance, tolerance errors in wheel track/wheelbase geometry (L, W) , and structural frame misalignments.
- **Integration and Drift Noise:** Computing global position via numerical integration of encoder-derived velocities over discrete sampling intervals dt leads to monotonic error accumulation (drift) over time.

2.5.3 Structure and Tuning Methodology of Matrix Q for Mecanum wheeled Platforms

Assuming a 2D pose state vector $x = [p_x \ p_y \ \theta]^\top$, the process noise covariance matrix Q_k is conventionally formulated as a diagonal matrix:

$$Q_k = \begin{bmatrix} \sigma_x^2 & 0 & 0 \\ 0 & \sigma_y^2 & 0 \\ 0 & 0 & \sigma_\theta^2 \end{bmatrix} \quad (2.18)$$

- **Position Components (σ_x^2, σ_y^2) :** Variance of position uncertainty along the longitudinal (X) and lateral (Y) axes. Due to severe lateral wheel slip during strafing maneuvers on Mecanum drives, practical tuning typically enforces $\sigma_y^2 > \sigma_x^2$.
- **Heading Component (σ_θ^2) :** Variance of rotational heading uncertainty. Rotational motion introduces substantial velocity differentials across the four wheels, frequently inducing eccentric angular slip.

Gyroscope Z-Axis Noise Model

At each sampling time t_i , the measured angular velocity from the z-axis gyroscope, denoted as z_i , is modeled as follows:

$$z_i = \omega_{\text{true}}(t_i) + b(t_i) + n_i \quad (2.19)$$

where:

- $\omega_{\text{true}}(t_i)$ represents the true angular velocity of the robot about the z-axis (rad/s).
- $b(t_i)$ denotes the MEMS sensor bias (zero-rate offset), which exhibits slow time-varying behavior.
- n_i represents the random measurement noise, modeled as a zero-mean, stationary white noise process with constant variance R and independent across temporal samples the exact parameter requiring characterization.

2.5.4 Role of the Measurement Noise Covariance Matrix R in the Extended Kalman Filter

2.5.5 Heteroscedasticity in Visual Measurement Noise

Unlike conventional sensors with stationary noise characteristics, pose estimation derived from planar fiducial markers via vision systems exhibits variable variance depending on observation conditions, a statistical phenomenon known as *heteroscedasticity* (non-homogeneous variance), as opposed to *homoscedasticity* (homogeneous variance). The fundamental origin of this behavior stems from the geometry of perspective projection: a fixed sub-pixel detection error at the image corner stage translates to a significantly larger spatial position error at greater distances, since a single pixel represents a larger physical space as distance from the camera increases.

Beyond distance dependence, several independent studies [popescu2017, pentenrieder2006, adamek2023] have reported the phenomenon of *pose ambiguity*. This induces abnormally elevated noise levels when the marker plane approaches parallelism with the camera's image plane (near-normal viewing angles), caused by the underlying mathematical ambiguity in resolving exact orientation from near-orthographic projection geometry.

Implication: A static covariance matrix R cannot accurately capture the noise dynamics of an ArUco sensor across its entire operational envelope. Consequently, an adaptive noise model for R is required, continuously modulating its values according to the instantaneous observation conditions of each individual measurement.

CHAPTER 3

HARDWARE DESIGN

3.1 Overall System Architecture

3.1.1 Two-Layer Distributed Architecture

The system is organized into a two-layer distributed architecture with a clear separation between the real-time control layer and the high-level algorithm processing layer, as illustrated in Figure 4.1.

- **Lower Layer – ESP32 (Real-time Control Layer):** Responsible for all time-deterministic tasks: reading encoders via hardware pulse counters (PCNT) at a 10 ms cycle, reading the IMU via I2C, executing the four-wheel velocity PI controller, and outputting PWM signals via the LEDC module. This layer operates independently of the PC's presence-upon connection loss, an automatic watchdog stops the motors after 3 seconds.
- **Upper Layer – PC (High-level Processing Layer):** Handles computationally intensive tasks that do not require hard real-time constraints: receiving telemetry from the ESP32, detecting Aruco marker via the webcam, executing the 50 Hz EKF, making navigation decisions via the Mission Controller, and sending velocity commands back to the ESP32.
- **Inter-layer Communication:** Utilizes USB Serial at 115,200 baud. This high-speed selection ensures that a 100-byte telemetry packet requires only about 1.1 ms to transmit-significantly less than the 20 ms EKF cycle, thereby preventing any bottleneck in the control loop.

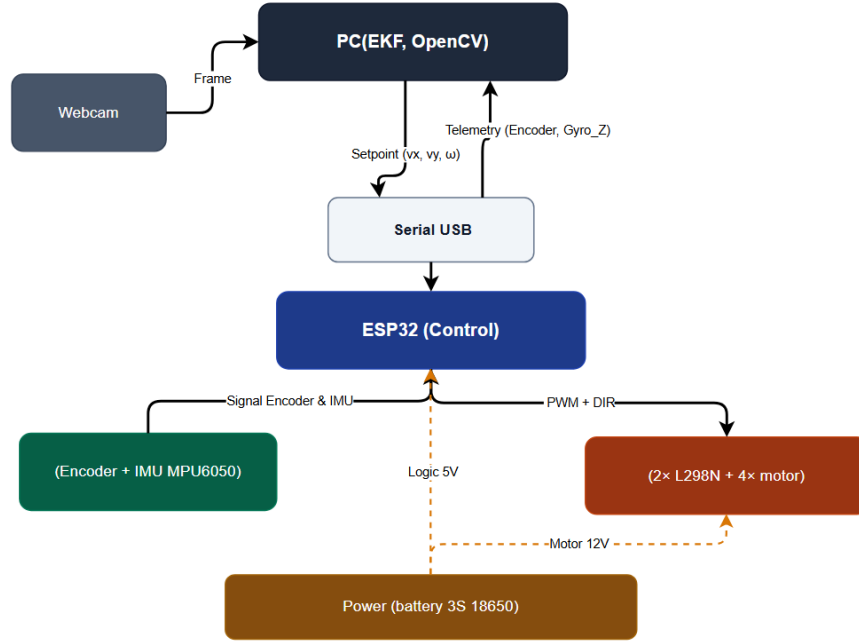


Figure 3.1: System architecture

The overall hardware and software architecture of the mobile robot system is structured into a two-tier hierarchical framework: a High-Level Computing Unit (PC) executing vision processing and state estimation, and a Low-Level Embedded Control Unit (ESP32) performing real-time motor control and sensor acquisition. The core functional blocks and signal flows are defined as follows:

- **Webcam Unit:** Captures raw image streams (Frame) and transfers them to the PC for computer vision processing via OpenCV.
- **High-Level Computing Unit (PC - EKF & OpenCV):** Acts as the main processing node. It performs visual fiducial marker tracking, executes the Extended Kalman Filter (EKF) for multi-sensor state estimation, and computes targeted velocity setpoints $\text{Setpoint } (v_x, v_y, w)$ sent to the lower tier.
- **Serial USB Interface:** Establishes a full-duplex communication channel between the PC and ESP32. It streams velocity commands down to the microcontroller and transmits sensor telemetry data (Encoder counts, Gyro Z) back to the PC.
- **Embedded Control Unit (ESP32):** Functions as the low-level real-time controller. It receives velocity setpoints, executes closed-loop motor PI

velocity control loops, processes raw sensor inputs, and generates control signals for the power drivers.

- **Sensor Feedback Block (Wheel Encoders + MPU-6050 IMU):** Measures individual wheel rotational speeds and the z-axis angular velocity (Gyro_Z), routing raw quadrature and I²C signals back to the ESP32.
- **Actuation System (2×L298N Drivers+4×DC Motors):** Receives Pulse Width Modulation (PWM) and directional (DIR) logic signals from the ESP32, amplifying power to drive the four Mecanum-wheeled DC motors.
- **Power Distribution Unit (3S 18650 Li-Ion Battery Pack):** Delivers isolated dual-voltage supply rails:
 - *5V Logic Power Rail:* Supplies stable power to the ESP32 and low-voltage logic microcircuits.
 - *12V Drive Power Rail:* Supplies high-current power directly to the L298N H-bridge motor drivers and DC motors.

3.1.2 Complete GPIO Pinout Assignment

The GPIO pinout was carefully designed to avoid pin strapping conflicts and to maximize the utilization of internal hardware peripherals:

Table 3.1: ESP32 GPIO pinout and hardware peripheral assignment

Device	Function	GPIO	HW Periph.	Notes
Motor FL	Forward PWM	25	LEDC ch 0	Output
Motor FL	Reverse PWM	13	LEDC ch 1	Output
Motor FR	Forward PWM	27	LEDC ch 2	Output
Motor FR	Reverse PWM	26	LEDC ch 3	Output
Motor RR	Forward PWM	23	LEDC ch 4	Output
Motor RR	Reverse PWM	16	LEDC ch 5	Output
Motor RL	Forward PWM	19	LEDC ch 6	Output
Motor RL	Reverse PWM	18	LEDC ch 7	Output
Encoder FL	Phase A/B	33, 32	PCNT unit 0	Internal pull-up
Encoder FR	Phase A/B	34, 35	PCNT unit 1	Input-only (*)
Encoder RR	Phase A/B	36, 39	PCNT unit 2	Input-only (*)
Encoder RL	Phase A/B	4, 17	PCNT unit 3	Internal pull-up
MPU-6050	SDA / SCL	21, 22	I2C0	4.7 k Ω external pull-up
USB Serial	TX / RX	1, 3	UART0	Via USB-UART bridge

3.2 Mecanum Robot

3.2.1 Mechanical Frame and Wheel Configuration

The robot is constructed on a rectangular aluminum profile frame, featuring four GA25-370 motors mounted at the four corners in a standard Mecanum configuration. The Front-Left (FL) and Rear-Right (RR) wheels are

equipped with rollers angled in the ”/” direction (when viewed from above), while the Front-Right (FR) and Rear-Left (RL) wheels have rollers angled in the ”\” direction, establishing a standard holonomic configuration.

The value of each components used are:

Table 3.2: Robot geometric parameters measured via vernier caliper

Parameter	Symbol	Measured Value	Importance in EKF
Mecanum wheel radius	R	34 mm	Linear/angular velocity ratio (FK, IK)
Half wheelbase (front-back)	l_x	100.5 mm	Coefficient in kinematics matrix
Half track width (left-right)	l_y	100 mm	Coefficient in kinematics matrix
Sum of half-dimensions	$L + W$	210.5 mm	Most critical parameter in IK/FK
Camera to robot center distance.	d_{cam}	[Measured] mm	Offset in frame conversion (Eq 2.37)

Measurement Technical Note: The parameter $(l_x + l_y) = 210.5$ mm was measured directly by summing the total distance from the left wheel center to the right wheel center, and the total distance from the front wheel center to the rear wheel center, then dividing the result by two, rather than adding the individual measurements of l_x and l_y . This method mitigates cumulative error and yields a highly accurate direct value for the Inverse/Forward Kinematics (IK/FK) equations.

3.2.2 Esp32-controller

Key Features for Application Suitability

The ESP32 (Espressif ESP32-WROOM-32) was selected for this project based on the following technical justifications:

- **Dual-core Xtensa LX6 240 MHz:** This architecture allows for the complete isolation of the PI control loop (running on Core 0) from serial communication and high-level tasks (running on Core 1). This separation prevents real-time jitter that could otherwise be caused by WiFi, Bluetooth, or communication interrupts.

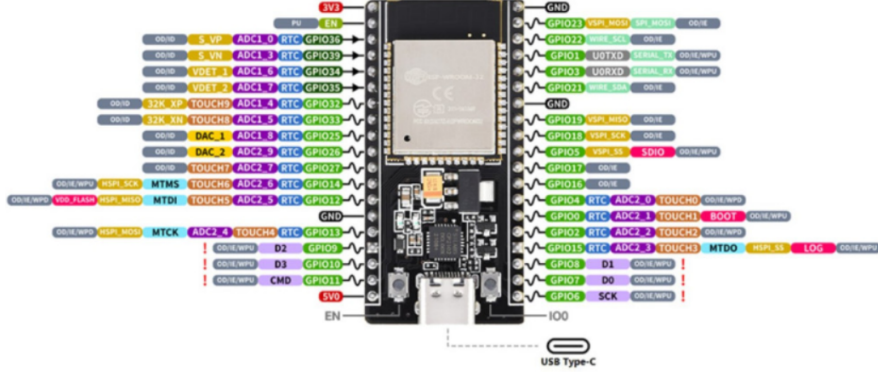


Figure 3.2: Esp32 devkit C

- **Hardware Pulse Counter (PCNT):** Four dedicated PCNT units are utilized to decode quadrature encoder signals directly via hardware. This ensures zero CPU overhead for pulse counting and guarantees that no pulses are missed, even at high motor speeds.
- **LEDC (LED Controller):** An 8-channel hardware PWM module is used to drive the motors. It offers flexible configuration for frequency and resolution, operating independently of the main software execution loops.
- **Hardware I2C at 400 kHz:** Provides stable and high-speed communication with the MPU-6050 IMU without the need for processor-intensive bit-banging.

3.2.3 GA25-370 Motors and Encoders

A set of four GA25-370 motors was selected based on an optimal balance between torque, speed, and precise encoder reading capabilities:

The encoder parameters, with $\text{CPR} = 4532$ and a wheel circumference of $2\pi R = 0.2136$ m, yield a spatial resolution of approximately $47 \mu\text{m}/\text{pulse}$, ensuring high precision for the odometry subsystem. Theoretically, the nominal no-load linear velocity calculated via:

$$v = \left(60 \text{ rpm} \times \frac{2\pi}{60} \right) \times 0.034 \approx 0.214 \text{ m/s} \quad (3.1)$$

compares with the experimentally measured FF_RPM_MAX of 49.57 rpm (yielding $v \approx 0.176$ m/s), showing an approximate 18% discrepancy. This margin is consistent with the voltage and mechanical losses through the L298N

Table 3.3: GA25-370 motor and encoder specifications

Parameter	Value	Notes
Nominal voltage	12 V DC	Supplied by 3S 11.1 V LiPo battery
No-load speed	60 RPM	At 12 V, without driver
Actual speed (via L298N)	~43 RPM	Due to ~2 V voltage drop
Gearbox ratio	1 : 103	Planetary gearbox
Encoder PPR (motor shaft)	11 PPR	Pre-gearbox, quadrature
CPR (post-gearbox, $\times 4$)	4,532 pulses/rev	$= 11 \times 4 \times 103$
Angular resolution	$0.0795^\circ/\text{pulse}$	$= 360^\circ/4,532$
Linear resolution	0.0465 mm/pulse	$= 2\pi R/4,532$

driver analyzed previously. The alignment between component selection and empirical findings establishes a coherent, systematic workflow: motor selection $\rightarrow$ driver selection $\rightarrow$ validated experimental outcomes.

3.2.4 L298N Motor Drivers

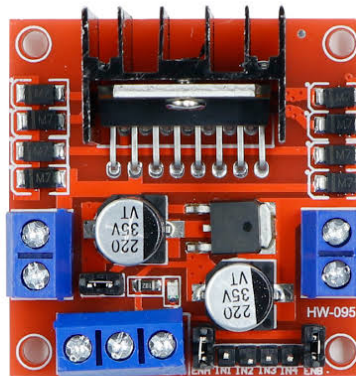


Figure 3.3: Motor driver L298n

Two L298N motor driver modules are utilized to control the four motors (each module controls two motors). The L298N is a BJT (Bipolar Junction

Transistor) H-bridge driver with specific characteristics that must be accounted for during firmware design:

- **Deadband:** The L298N exhibits a voltage drop of approximately 2 V across its transistors, meaning a minimum input voltage is required before the motor begins to rotate. The value $\text{PWM_MIN} = 65/255$ ($\approx 25.5\%$ duty cycle) was determined empirically to reliably overcome the deadband across all operating temperatures.
- **Actual motor voltage:** Given a 12 V supply and a ~ 2 V voltage drop, the motor receives approximately 10 V at 100% duty cycle-resulting in a speed reduction of $\sim 17\%$ compared to the nominal specification. The FF_RPM_MAX parameter was measured empirically for each individual motor to accurately compensate for this discrepancy in the feedforward control loop.
- **PWM Frequency:** A frequency of $f_{\text{PWM}} = 5,000$ Hz (5 kHz) was selected. This falls within an efficient operating range for the L298N-preventing overheating caused by excessively high-frequency switching (e.g., > 20 kHz) while adequately managing electromagnetic noise considerations.

3.2.5 Inertial Measurement Unit: MPU-6050

The MPU-6050 was selected as the primary Inertial Measurement Unit (IMU) for the mobile robot system due to its integrated 6-degree-of-freedom (6-DOF) architecture-incorporating a 3-axis accelerometer and a 3-axis gyroscope on a single monolithic MEMS die. Its hardware interface relies on the standard I²C communication bus, seamlessly integrating with the 10 ms sensor sampling task running on the ESP32 microcontroller. Additionally, its low cost, compact form factor, and robust community software support make it highly suitable for embedded robotics deployment.



Figure 3.4: MPU-6050

Key Technical Specifications

Table 3.4 details the core electrical and kinematic specifications of the MPU-6050 sensor module.

Table 3.4: MPU-6050 Hardware and Electrical Specifications

Specification / Parameter	Configured Value / Range
Communication Interface	I ² C (Standard Mode: 100 kHz / Fast Mode: 400 kHz)
I ² C Bus Address	0x68 (AD0 → GND) or 0x69 (AD0 → VCC)
Programmable Accelerometer Range	$\pm 2g$, $\pm 4g$, $\pm 8g$, $\pm 16g$
Programmable Gyroscope Range	$\pm 250^\circ/\text{s}$, $\pm 500^\circ/\text{s}$, $\pm 1000^\circ/\text{s}$, $\pm 2000^\circ/\text{s}$
Analog-to-Digital Converter (ADC)	16-bit resolution per channel (Accel and Gyro)
Operating Supply Voltage	2.375 V–3.46 V (3.3 V logic compatible with ESP32)
Digital Low-Pass Filter (DLPF)	Programmable bandwidth vs. group delay trade-off

Experimental Hardware Configuration

The specific hardware parameters and ESP32 pin assignments used in the prototype platform are summarized in Table 3.5.

Table 3.5: System Configuration Parameters for MPU-6050

Parameter	Selected Configuration
Accelerometer Full-Scale Range	$[\pm 4g]$
Gyroscope Full-Scale Range	$[\pm 250^\circ/s]$
Sampling Rate / Output Rate	$[100\text{ Hz} / 10\text{ ms interval}]$
Digital Low-Pass Filter (DLPF_CFG)	$[Config\ 2\ (21\text{ Hz Bandwidth},\ 2\text{ ms Delay})]$
ESP32 Pin Assignments	SDA $\rightarrow$ GPIO21, SCL $\rightarrow$ GPIO22
I ² C Clock Frequency	$[400\text{ kHz}]$

Note on Mathematical Modeling: The detailed statistical noise models of the accelerometer and gyroscope-including zero-mean Gaussian measurement noise and slow time-varying bias drift terms (b_a, b_g). Consequently, this section strictly focuses on hardware characteristics, physical interface pinouts, and selection rationale to avoid content redundancy.

3.2.6 Role in the Mecanum Robot’s Localization System

In this project, data from the MPU6050 is utilized to address two primary tasks:

- **Heading Estimation:** Integrating the Z-axis angular velocity (g_z or ω_z) to determine the robot’s Yaw angle. This is particularly crucial because Mecanum wheels frequently experience slippage, rendering encoder data unreliable for orientation tracking over time.

3.2.7 Solutions for Enhancing Signal Stability

Due to the specific mechanical characteristics of the Mecanum robot, which generate significant vibrations from the rollers and magnetic interference from the four DC motors, the system employs both hardware- and software-level filtering solutions:

- **Hardware Filtering:** Utilizing parallel capacitors ($100\ \mu\text{F}$ and $0.1\ \mu\text{F}$) positioned closely to the sensor’s power pins to prevent voltage drops and filter out high-frequency electrical noise.

USB Webcam



Figure 3.5: Usb-webcam

The system utilizes a UGREEN 2K USB Webcam, which is securely fixed to the robot frame, facing forward along the robot’s $+X$ axis. The camera connects to the PC via a USB 2.0 port, operating at a resolution of 640×480 pixels at 30 fps.

Mounting Position: The camera is positioned at a distance d_{cam} ahead of the robot’s geometric center. This creates a camera offset that is compensated for during coordinate frame conversion (Equation 2.37). The camera is mounted at a height such that the optical center of the lens is approximately level with the center of the wall-mounted Aruco marker. This alignment minimizes the t_y error component during pose estimation.

3.3 Motor Control System

Velocity Data Acquisition (PCNT)

The Hardware Pulse Counter (PCNT) module of the ESP32 is utilized to count encoder pulses in Quadrature $\times 4$ mode. This hardware-level counting ensures high resolution while minimizing CPU overhead.

The pulse delta calculation ($dt = 10$ ms) is defined as:

$$\Delta \text{ticks}_i(k) = \text{count}_i(k) - \text{count}_i(k - 1) \quad (3.2)$$

The actual velocity in Revolutions Per Minute (RPM) is calculated by:

$$n_i = \frac{60 \cdot \Delta \text{ticks}_i}{\text{CPR} \cdot dt} \quad [\text{RPM}] \quad (3.3)$$

Given a $\text{CPR} = 4532$, at a speed of 43 RPM, $\Delta \text{ticks} \approx 32$, which remains well within the capacity of the 16-bit hardware counter.

Exponential Moving Average (EMA) Filter

To mitigate quantization noise at low velocities, an Exponential Moving Average (EMA) filter is applied:

$$\hat{n}_i(k) = \alpha \cdot n_i(k) + (1 - \alpha) \cdot \hat{n}_i(k - 1) \quad (3.4)$$

With $\alpha = 0.15$, the resulting time constant $\tau \approx 63$ ms ensures a smooth signal while remaining significantly faster than the motor's mechanical time constant (200 ms).

PI Controller and Feedforward

The control architecture combines Feedforward (FF) for static compensation and a PI controller for dynamic regulation.

1. **Feedforward Component:** This compensates for the motor deadband and voltage drops across the L298N driver.

$$u_{FF,i} = \frac{|\omega_{sp,i}|}{n_{max,i}} \cdot (u_{max} - u_{min}) + \text{sign}(\omega_{sp,i}) \cdot u_{min} \quad (3.5)$$

Parameters: $u_{max} = 250$, $u_{min} = 65$. The $n_{max,i}$ value is calibrated individually for each motor to compensate for manufacturing tolerances.

2. **PI Component:** Designed to eliminate the steady-state error $e_i(k) = \omega_{sp,i} - \hat{n}_i(k)$.

$$u_{PI,i}(k) = K_{p,i}e_i(k) + K_{i,i} \sum e_i(k)dt \quad (3.6)$$

3. **Output Summation and Anti-windup:** The final PWM signal is clamped within the range $[-250, 250]$. To prevent integral windup, which causes saturation and loss of control, the integral term is limited to $|\sum e_i dt| \leq 30$.

3.4 ArUco Marker Deployment Strategy

Fiducial Placement Guidelines and Technical Requirements

ArUco fiducial markers serve as stationary exteroceptive landmarks providing absolute pose observations for the Extended Kalman Filter (EKF) localization pipeline. Marker positions can be deployed arbitrarily across the operational environment without requiring rigid grid topologies or uniform spatial intervals. Global landmark coordinates (X_m, Y_m, θ_m) are logged automatically via an offline calibration script relative to the initial robot reference pose $(0, 0, 0)$.

Table 3.6 details the hardware mounting constraints and technical criteria established for marker installation.

Table 3.6: ArUco Marker Installation Guidelines and Technical Rationale

Installation Criterion	Technical Requirement	Engineering Rationale
Mounting Location	Arbitrary fixed wall/column surfaces	Accommodates physical site topology
Sensing Distance (d)	$0.5 \text{ m} \leq d \leq 2.5 \text{ m}$ to camera	Maintains image resolution; avoids FOV clipping
Tilt Incident Angle	Angular offset $\leq 45^\circ$ to optical axis	Suppresses perspective distortion & pose jitter
Surface Flatness	Planar mounting; anti-glare finish	Prevents 3D pose degradation & corner noise
Landmark Density	Visibility ≥ 1 marker along trajectory	Enables periodic EKF drift correction

3.4.1 Mapping Data Processing and Landmark Initialization

During the environment mapping phase, raw visual readings captured while the robot remains stationary at the origin are processed to initialize landmark coordinates (X_m, Y_m, θ_m) in the map frame:

1. **Coordinate Frame Conversion:** Raw local measurements in the Camera Optical Frame $(x_{\text{local}}, z_{\text{local}})$ are mapped to the ROS World

Frame (REP-105 standard):

$$X_m = d_{\text{offset}} + z_{\text{local}} \quad (\text{m}) \quad (3.7)$$

$$Y_m = -x_{\text{local}} \quad (\text{m}) \quad (3.8)$$

where $d_{\text{offset}} = 0.13$ m represents the forward camera lens offset relative to the robot center of rotation.

2. **Temporal Noise Filtering and Circular Averaging:** To eliminate sub-pixel corner detection noise and quantization jitter during static logging, spatial coordinates (X_m, Y_m) are computed via arithmetic sample averaging across N continuous frames:

$$\bar{X}_m = \frac{1}{N} \sum_{i=1}^N X_{m,i}, \quad \bar{Y}_m = \frac{1}{N} \sum_{i=1}^N Y_{m,i} \quad (3.9)$$

Orientation θ_m is resolved using trigonometric vector summation (circular mean) to prevent phase wrap discontinuities around $\pm\pi$:

$$\bar{\theta}_m = \text{atan2} \left(\frac{1}{N} \sum_{i=1}^N \sin \theta_{m,i}, \frac{1}{N} \sum_{i=1}^N \cos \theta_{m,i} \right) \quad (3.10)$$

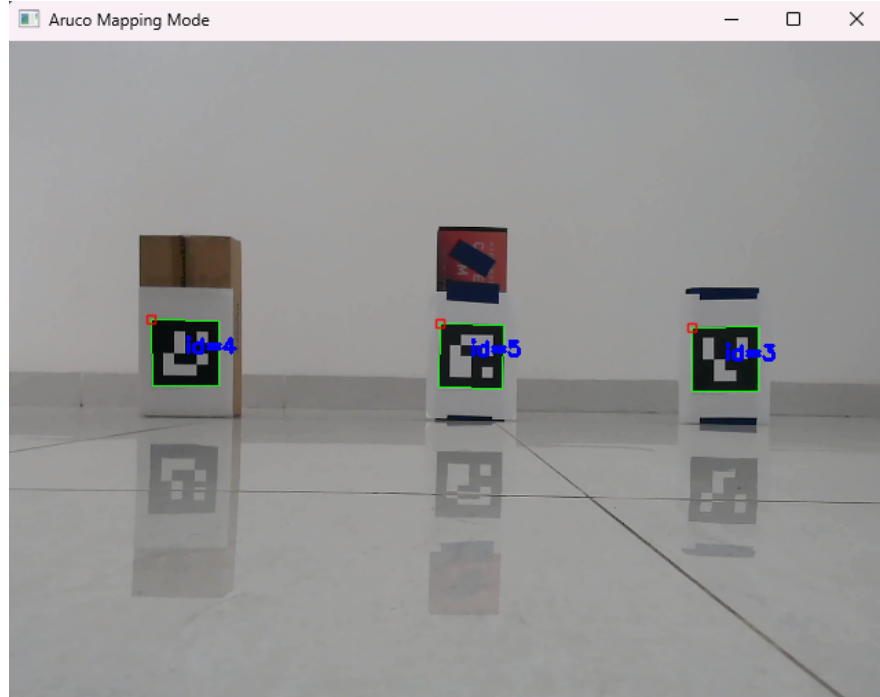


Figure 3.6: Marker layout

CHAPTER 4

SOFTWARE DESIGN

4.1 Communication esp32 and computer

The system software is organized into two distinct layers that are completely independent in terms of programming language, execution environment, and operational duty cycles. These layers communicate via a predefined text-based protocol.

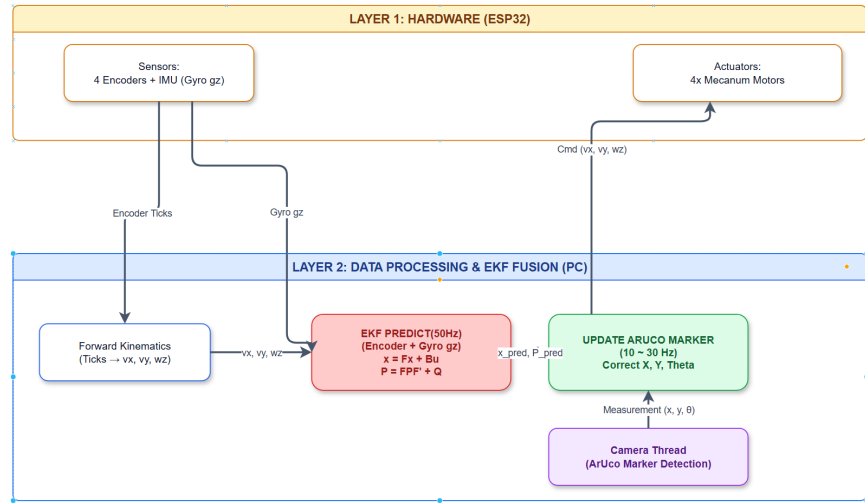


Figure 4.1: Two-layer distributed architecture block diagram

4.1.1 Telemetry Protocol

The telemetry protocol is designed to transmit sufficient data for EKF computation while minimizing bandwidth overhead. Packets are transmitted every 20 ms (50 Hz).

4.1.2 Camera Calibration

Objective and Theoretical Method

The primary goal of the camera calibration procedure is to determine the intrinsic parameter matrix K and the lens distortion coefficient vector D . Estimating these parameters enables digital image undistortion and provides accurate optical geometry required for solving the Perspective- n -Point (PnP) problem during ArUco marker pose estimation. The calibration pipeline implements Zhang’s planar pattern method [zhang2000] combined with the Brown-Conrady radial-tangential lens distortion model.

Experimental Setup and Implementation

The empirical setup and software architecture for calibration were established as follows:

- **Calibration Target:** An 11×8 planar checkerboard pattern containing 10×7 internal grid corners with a square grid dimension of $a = 20$ mm.
- **Optical Sensor:** Monocular camera operating at a fixed resolution of 640×480 pixels.
- **Dataset Acquisition:** A total of 70 valid image frames were captured under diverse tilt angles, distances, and grid orientations across the camera’s field of view.
- **Detection Pipeline:** A two-stage Python implementation was executed: a fast initial detection pass operating on $\frac{1}{8}$ downsampled previews, followed by sub-pixel corner refinement using OpenCV’s `cv2.cornerSubPix` with an 11×11 search window and a maximum iteration threshold of 30 cycles (or convergence precision $\epsilon = 0.001$).

Calibration Results and Evaluation

The estimated camera intrinsic matrix K is expressed as:

$$K = \begin{bmatrix} 564.33 & 0 & 320.81 \\ 0 & 564.27 & 245.76 \\ 0 & 0 & 1 \end{bmatrix} \quad (4.1)$$

The estimated distortion vector $D = [k_1, k_2, p_1, p_2, k_3]$ modeling radial and tangential lens distortion parameters is given by:

$$D = [0.10218 \quad 0.03381 \quad -0.00447 \quad 0.00007 \quad -0.36132] \quad (4.2)$$

Table 4.1 provides a quantitative assessment of the key estimated parameters relative to theoretical baseline metrics.

Table 4.1: Camera Calibration Parameter Evaluation

Parameter / Metric	Empirical Value	Technical Evaluation
Valid Sample Size	70 frames	Sufficient view diversity (> 10 frames threshold)
Focal Length (f_x, f_y)	564.33 px, 564.27 px	Ideal 1 : 1 square pixel aspect ratio
Principal Point (c_x, c_y)	(320.81, 245.76) px	< 1% offset relative to optical center (320, 240)
Mean Reprojection Error	0.502174 pixels	Highly accurate (< 1.0 px benchmark for mobile robotics)

Summary

The calibration yields an overall mean RMS reprojection error of 0.502 pixels. This low reprojection error confirms that the derived intrinsic matrix K and distortion coefficients D provide sufficient geometric accuracy for online image undistortion and subsequent ArUco fiducial marker pose estimation.

4.2 Aruco marker detection

4.2.1 System Configuration and Symbol Nomenclature

The computer vision module receives raw video frames (640×480 px) from the onboard camera, decodes planar ArUco fiducial markers, and computes the absolute spatial pose of the mobile robot within the World Coordinate Frame $\{W\}$ adhering to the ROS REP-105 standard (+X Forward, +Y Left). Table 4.2 details the mathematical symbols, physical units, and reference frames utilized throughout the visual localization pipeline.

Table 4.2: Vision Subsystem Parameter Nomenclature and Reference Frames

Symbol	Value / Unit	Description	Reference Frame
s	0.15 m (15 cm)	Outer physical side length of ArUco marker	$\{M\}$ (Marker)
d_{offset}	0.13 m (13 cm)	Longitudinal offset from camera lens to robot center	$\{R\}$ (Robot)
K, D	$3 \times 3, 1 \times 5$	Camera intrinsic matrix and distortion vector	$\{C\}$ (Camera)
x_m, y_m, θ_m	Static (m, rad)	Global pose coordinates of marker in map database	$\{W\}$ (World)
$x_{\text{local}}, z_{\text{local}}$	Dynamic (m)	Lateral (X) offset and depth (Z) distance to marker	$\{C\}$ (Camera)
$\text{yaw}_{\text{local}}$	Dynamic (rad)	Relative yaw angle between camera optical axis and marker	$\{C\}$ (Camera)
x_c, y_c, θ_c	Dynamic (m, rad)	Transformed absolute position and heading of camera lens	$\{W\}$ (World)
x_r, y_r, θ_r	Dynamic (m, rad)	Computed absolute pose of robot center of rotation	$\{W\}$ (World)
z_k	Vector $\mathbb{R}^3$	Measurement observation vector feeding the EKF update	$\{W\}$ (World)

4.2.2 Stage 1: Local Pose Estimation

The Stage A1 execution pipeline determines the relative spatial transformation between the detected marker and the camera origin within

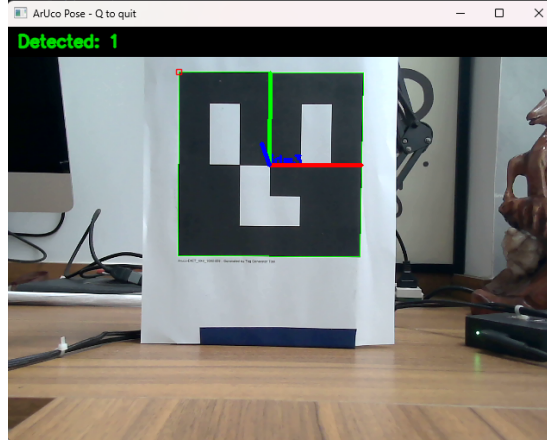


Figure 4.2: Aruco marker detection

the OpenCV Optical Frame $\{C\}$ (+Z Depth, +X Right, +Y Down):

1. **Fiducial Corner Extraction:** Converts input color frames to grayscale and executes `cv2.aruco.ArucoDetector` using the `DICT_4X4_100` dictionary to extract sub-pixel 2D image coordinates for all four marker corners.
2. **3D Perspective- n -Point (PnP) Resolution:** Solves the geometric PnP problem using `cv2.solvePnPGeneric` configured with the `SOLVEPNP_IPPE_SQUARE` solver flag. Designed specifically for planar square targets ($s = 0.15$ m), this solver resolves rotational ambiguity (Pose Ambiguity) and selects the candidate pose minimizing reprojection error.
3. **Local Coordinate Extraction:**

$$x_{\text{local}} = \text{tvec}[0] \quad (\text{m}) \quad (4.3)$$

$$z_{\text{local}} = \text{tvec}[2] \quad (\text{m}) \quad (4.4)$$

$$\text{yaw}_{\text{local}} = -\text{atan2}(-R_{0,2}, R_{0,0}) \quad (\text{rad}) \quad (4.5)$$

where $R \in \mathbb{SO}(3)$ is the 3×3 rotation matrix converted from the Rodrigues vector `rvec`.

4.2.3 Mathematical Foundation of $SE(3)$ Homogeneous Transformations and Stage 2 Mapping

The continuous spatial transformation chain mapping coordinates from the Marker frame $\{M\}$, to the Camera optical frame $\{C\}$, Robot body frame

$\{R\}$, and global World frame $\{W\}$ is generalized via Special Euclidean group $SE(3)$ homogeneous matrix multiplication:

$$T_{W \leftarrow R} = T_{W \leftarrow M} \cdot (T_{C \leftarrow M})^{-1} \cdot (T_{R \leftarrow C})^{-1} \quad (4.6)$$

where $T_{R \leftarrow C} \in SE(3)$ incorporates the physical forward mounting offset $d_{\text{offset}} = 0.13$ m. Projecting this 3D spatial transformation onto planar 2D space yields the analytical real-time equations evaluated at each execution step:

Absolute Camera Heading in World Frame (θ_c):

$$\theta_c = \theta_m - \text{yaw}_{\text{local}} \quad (\text{rad}) \quad (4.7)$$

Camera Lens Position in World Frame (x_c, y_c):

$$x_c = x_m - (z_{\text{local}} \cdot \cos \theta_c + x_{\text{local}} \cdot \sin \theta_c) \quad (\text{m}) \quad (4.8)$$

$$y_c = y_m - (z_{\text{local}} \cdot \sin \theta_c - x_{\text{local}} \cdot \cos \theta_c) \quad (\text{m}) \quad (4.9)$$

Robot Center Offset Compensation (x_r, y_r, θ_r):

$$x_r = x_c - d_{\text{offset}} \cdot \cos \theta_c \quad (\text{m}) \quad (4.10)$$

$$y_r = y_c - d_{\text{offset}} \cdot \sin \theta_c \quad (\text{m}) \quad (4.11)$$

$$\theta_r = \theta_c \quad (\text{rad}) \quad (4.12)$$

Observation Vector Packaging for EKF Update

The resolved absolute coordinates (x_r, y_r, θ_r) corresponding to the robot's center of rotation are structured into the exteroceptive measurement observation vector $z_k \in \mathbb{R}^3$:

$$z_k = \begin{bmatrix} x_{\text{aruco}} \\ y_{\text{aruco}} \\ \theta_{\text{aruco}} \end{bmatrix} = \begin{bmatrix} x_r \\ y_r \\ \theta_r \end{bmatrix} \quad (4.13)$$

The vector z_k is directly injected into the Update phase of the Extended Kalman Filter to bound the accumulated odometry drift whenever an ArUco fiducial target enters the visual field of view.

4.3 Extended Kalman Filter (EKF) Algorithm Architecture

4.3.1 State Representation and General Framework

The mobile robot state at discrete time step t is represented by the 2D pose vector $x_t \in \mathbb{R}^3$ in the global map coordinate frame (REP-105 standard):

$$x_t = \begin{bmatrix} p_{x,t} \\ p_{y,t} \\ \theta_t \end{bmatrix} \quad (4.14)$$

where $p_{x,t}$ and $p_{y,t}$ denote global Cartesian positions, and $\theta_t \in (-\pi, \pi]$ represents the absolute heading angle.

The non-linear system dynamics and exteroceptive measurement observations are governed by the general discrete-time state-space formulation:

$$x_t = f(x_{t-1}, u_t) + w_t, \quad w_t \sim \mathcal{N}(0, Q_t) \quad (4.15)$$

$$z_t = h(x_t) + v_t, \quad v_t \sim \mathcal{N}(0, R_t) \quad (4.16)$$

where w_t and v_t denote mutually uncorrelated zero-mean white Gaussian process and measurement noise vectors with covariance matrices $Q_t \in \mathbb{R}^{3 \times 3}$ and $R_t \in \mathbb{R}^{3 \times 3}$, respectively.

The recursive estimation loop proceeds via two alternating phases:

- **Prediction Step (Prior State Propagation):**

$$\bar{x}_t = f(\hat{x}_{t-1}, u_t) \quad (4.17)$$

$$\bar{P}_t = F_t P_{t-1} F_t^\top + Q_t \quad (4.18)$$

- **Correction Step (Posterior State Update):**

$$K_t = \bar{P}_t H_t^\top (H_t \bar{P}_t H_t^\top + R_t)^{-1} \quad (4.19)$$

$$x_t = \bar{x}_t + K_t (z_t - h(\bar{x}_t)) \quad (4.20)$$

$$P_t = (I - K_t H_t) \bar{P}_t \quad (4.21)$$

4.3.2 Prediction Stage Implementation

Control Vector Specification

The system control input vector $u_t \in \mathbb{R}^3$ comprises body-frame linear translational velocities and an effective rotational yaw rate:

$$u_t = \begin{bmatrix} v_{x,u} \\ v_{y,u} \\ \omega_{\text{eff}} \end{bmatrix} \quad (4.22)$$

where $v_{x,u}$ and $v_{y,u}$ represent robot body longitudinal and lateral velocities derived from wheel encoder quadrature counts via Mecanum forward kinematics. The scalar ω_{eff} represents the effective angular velocity, dynamically selected based on IMU sensor health:

$$\omega_{\text{eff}} = \begin{cases} g_z & \text{if imu_valid} = 1 \text{ (MPU-6050 Gyroscope Z-axis)} \\ \omega_{\text{encoder}} & \text{if imu_valid} = 0 \text{ (Kinematic Wheel Encoder Fallback)} \end{cases} \quad (4.23)$$

Non-Linear State Transition Model $f(x, u)$

The discretization of continuous kinematic equations over the sampling interval Δt produces the prior state prediction model:

$$p_{x,t} = p_{x,t-1} + (v_{x,u} \cos \theta_{t-1} - v_{y,u} \sin \theta_{t-1}) \Delta t \quad (4.24)$$

$$p_{y,t} = p_{y,t-1} + (v_{x,u} \sin \theta_{t-1} + v_{y,u} \cos \theta_{t-1}) \Delta t \quad (4.25)$$

$$\theta_t = \text{wrapToPi}(\theta_{t-1} + \omega_{\text{eff}} \Delta t) \quad (4.26)$$

where $\text{wrapToPi}(\cdot)$ constrains the updated orientation within the interval $(-\pi, \pi]$.

Jacobian Matrices F_t and G_t

Linearizing the non-linear transition function $f(x, u)$ requires evaluating first-order partial derivative Jacobians at the prior posterior state estimate $\hat{x}_{t-1}$ and control input u_t :

State Jacobian Matrix $F_t = \left. \frac{\partial f}{\partial x} \right|_{\hat{x}_{t-1}, u_t} \in \mathbb{R}^{3 \times 3}$:

$$F_t = \begin{bmatrix} 1 & 0 & (-v_{x,u} \sin \theta_{t-1} - v_{y,u} \cos \theta_{t-1}) \Delta t \\ 0 & 1 & (v_{x,u} \cos \theta_{t-1} - v_{y,u} \sin \theta_{t-1}) \Delta t \\ 0 & 0 & 1 \end{bmatrix} \quad (4.27)$$

Control Jacobian Matrix $G_t = \left. \frac{\partial f}{\partial u} \right|_{\hat{x}_{t-1}, u_t} \in \mathbb{R}^{3 \times 3}$:

$$G_t = \begin{bmatrix} \cos \theta_{t-1} \cdot \Delta t & -\sin \theta_{t-1} \cdot \Delta t & 0 \\ \sin \theta_{t-1} \cdot \Delta t & \cos \theta_{t-1} \cdot \Delta t & 0 \\ 0 & 0 & \Delta t \end{bmatrix} \quad (4.28)$$

Process Noise Covariance Mapping Q_t

The dynamic state process noise matrix Q_t is mapped from the control space noise matrix Q_u using Jacobian G_t :

$$Q_t = G_t Q_u G_t^\top \quad \text{with} \quad Q_u = \text{diag} \left(\sigma_{v_x}^2, \sigma_{v_y}^2, q_\omega \right) \quad (4.29)$$

where control variance terms $\sigma_{v_x}^2$, $\sigma_{v_y}^2$, and q_ω are updated online based on control command magnitudes.

Prediction Phase Summary

The complete prior state and covariance propagation outputs are expressed as:

$$\bar{x}_t = f(\hat{x}_{t-1}, u_t) \quad (4.30)$$

$$\bar{P}_t = F_t P_{t-1} F_t^\top + Q_t \quad (4.31)$$

4.3.3 Correction Stage Implementation

ArUco Measurement Pre-Transformation

Raw optical observations $(x_{\text{local}}, z_{\text{local}}, \text{yaw}_{\text{local}})$ captured in the camera frame $\{C\}$ (REP-103) are pre-transformed into the global map frame $\{W\}$ (REP-105) prior to filter injection. Given global marker coordinates (X_m, Y_m, θ_m) and forward camera lens offset $d = 0.13$ m:

Stage 1 (Global Camera Pose Integration):

$$\theta_c = \theta_m + \text{yaw}_{\text{local}} - \pi \quad (4.32)$$

$$X_c = X_m - x_{\text{local}} \sin \theta_c - z_{\text{local}} \cos \theta_c \quad (4.33)$$

$$Y_c = Y_m + x_{\text{local}} \cos \theta_c - z_{\text{local}} \sin \theta_c \quad (4.34)$$

Stage 2 (Robot Center Offset Compensation):

$$\theta_r = \theta_c \quad (4.35)$$

$$X_r = X_c - d \cos \theta_c \quad (4.36)$$

$$Y_r = Y_c - d \sin \theta_c \quad (4.37)$$

This two-stage transformation yields the absolute measurement vector $z_t = [X_r, Y_r, \theta_r]^\top$, matching the dimensions and frame of state vector x_t .

Measurement Model $h(x)$ and Static Observation Matrix H

Because input measurement vector z_t is transformed into global state space units prior to filtering, the internal EKF measurement function reduces to an identity mapping:

$$h(x_t) = x_t \implies H = \frac{\partial h}{\partial x} = I_3 \in \mathbb{R}^{3 \times 3} \quad (4.38)$$

Using a constant identity matrix $H = I_3$ eliminates the need to linearize measurement equations online during update cycles.

Global Measurement Noise Covariance R_t

Local visual noise R_{local} is propagated to global map coordinates via transformation Jacobians J_1 and J_2 :

$$R_t = J_2 \left(J_1 R_{\text{local}} J_1^\top \right) J_2^\top \quad (4.39)$$

where transformation Jacobians J_1 and J_2 are defined as:

$$J_1 = \begin{bmatrix} -\sin \theta_c & -\cos \theta_c & -x_{\text{local}} \cos \theta_c + z_{\text{local}} \sin \theta_c \\ \cos \theta_c & -\sin \theta_c & -x_{\text{local}} \sin \theta_c - z_{\text{local}} \cos \theta_c \\ 0 & 0 & 1 \end{bmatrix} \quad (4.40)$$

$$J_2 = \begin{bmatrix} 1 & 0 & d \sin \theta_c \\ 0 & 1 & -d \cos \theta_c \\ 0 & 0 & 1 \end{bmatrix} \quad (4.41)$$

Correction Equations

Substituting $H = I_3$ simplifies the standard update formulations:

$$K_t = \bar{P}_t (\bar{P}_t + R_t)^{-1} \quad (4.42)$$

$$x_t = \bar{x}_t + K_t \cdot \text{wrapToPi}(z_t - \bar{x}_t) \quad (4.43)$$

$$P_t = (I - K_t) \bar{P}_t \quad (4.44)$$

where angular innovation subtraction $(z_{t,\theta} - \bar{x}_{t,\theta})$ is wrapped within $(-\pi, \pi]$.

4.3.4 Full Iterative Execution Loop

The recursive flow of the EKF localization pipeline is structured as follows:

$$(x_{t-1}, P_{t-1}) \xrightarrow[\text{Control } u_t]{\text{Prediction}} (\bar{x}_t, \bar{P}_t) \xrightarrow[\text{Measurement } (z_t, R_t)]{\text{Correction}} (x_t, P_t) \quad (4.45)$$

The posterior state estimate x_t and error covariance P_t serve as prior inputs for the subsequent prediction cycle $t + 1$.

4.3.5 Normalized Innovation Squared (NIS) Verification and Outlier Gating

4.3.6 Theoretical Formulation and Gating Mechanism

Prior to executing the Kalman measurement update step, each incoming exteroceptive ArUco visual observation is subjected to statistical validation using the Normalized Innovation Squared (NIS) metric ϵ_k :

$$\epsilon_k = \nu_k^\top S_k^{-1} \nu_k \quad (4.46)$$

where the measurement innovation vector $\nu_k \in \mathbb{R}^3$ and the innovation covariance matrix $S_k \in \mathbb{R}^{3 \times 3}$ are formulated as:

$$\nu_k = z_k - H\hat{x}_k^- \quad (4.47)$$

$$S_k = HP_k^- H^\top + R_k \quad (4.48)$$

Here, ν_k represents the residual vector evaluating the spatial discrepancy between the pre-transformed visual observation z_k and the prior state prediction $\hat{x}_k^-$, whereas S_k denotes the total projected uncertainty in the measurement space.

Assuming that the state error covariance P_k^- and the measurement noise matrix R_k are correctly calibrated, the scalar NIS metric ϵ_k obeys a chi-squared (χ^2) distribution with $m = 3$ degrees of freedom, corresponding to the global state coordinates $[x, y, \theta]^\top$:

$$\epsilon_k \sim \chi_m^2, \quad m = 3 \quad (4.49)$$

To safeguard the filter against spurious visual detections, misidentifications, or PnP pose estimation spikes, a Chi-Square Gating mechanism is enforced. A visual observation is admitted into the update step if and only if the calculated NIS metric satisfies the 95% statistical confidence threshold:

$$\epsilon_k \leq \chi^2(3, 0.95) = 7.815 \quad (4.50)$$

If $\epsilon_k > 7.815$, the measurement is flagged as a statistical outlier and rejected. Upon rejection, the filter bypasses the update stage, directly retaining the prior state $\hat{x}_k^-$ and prior covariance P_k^- as posterior outputs for the current cycle, thereby preventing state corruption from invalid visual observations.

4.4 Systematic Estimation of Process Noise Covariance Matrix Q_{odom} for Mecanum Wheel Odometry

4.4.1 Objective and Kinematic Mapping

The primary objective is to identify the control noise covariance matrix $R_u = \text{diag}(\sigma_{v_x}^2, \sigma_{v_y}^2)$, derived from wheel encoder pulses mapped through Mecanum forward kinematics. In the prediction step of the Extended Kalman Filter (EKF), the state noise covariance matrix $Q_{\text{odom},k}$ is dynamically computed via linear error propagation:

$$Q_{\text{odom},k} = G_k R_{u,k} G_k^\top \quad (4.51)$$

where G_k represents the kinematic Jacobian mapping local linear control velocity noise into global state pose uncertainty.

4.4.2 Velocity-Dependent Step Noise Model

At each discrete sampling interval Δt , the single-step velocity variance $\sigma_{\text{step}}^2(v)$ is modeled as a composite function of stationary and speed-proportional disturbance sources:

$$\sigma_{\text{step}}^2(v) = \sigma_{\text{floor}}^2 + \delta \cdot |v| \quad (4.52)$$

where:

- σ_{floor}^2 : Velocity-independent noise floor arising from encoder quantization limits, mechanical backlash, and shaft bearing friction.
- $\delta \cdot |v|$: Velocity-proportional variance accounting for dynamic wheel slip, roller contact point compliance, and floor surface variations.

The model requires identifying two scalar parameters for each motion axis: the offset parameter σ_{floor}^2 and the proportional slip coefficient δ .

4.4.3 Experimental Protocol and Total Variance Derivation

To isolate noise channels, pure-axis translation trials are executed along independent kinematic axes (v_x, v_y, ω_z). A fixed trajectory distance $D = 1.0$ m is maintained across all test runs while sweeping across multiple nominal velocities v .

For a given velocity v , the total discrete sampling steps required to traverse distance D is given by:

$$N(v) = \frac{D}{v \cdot \Delta t} \quad (4.53)$$

Assuming noise uncorrelation across sequential sampling steps, the cumulative total position variance $\text{Var}_{\text{total}}(v)$ accumulated at the end of trajectory D expands as:

$$\begin{aligned}\text{Var}_{\text{total}}(v) &= N(v) \cdot \sigma_{\text{step}}^2(v) \cdot \Delta t^2 \\ &= \left(\frac{D}{v \cdot \Delta t} \right) (\sigma_{\text{floor}}^2 + \delta \cdot v) \Delta t^2 \\ &= \delta \cdot D \cdot \Delta t + \sigma_{\text{floor}}^2 \cdot D \cdot \Delta t \cdot \frac{1}{v}\end{aligned}\quad (4.54)$$

4.4.4 Single-Variable Linear Regression Formulation

Although the physical noise model contains two unknown parameters $(\delta, \sigma_{\text{floor}}^2)$, the experimental identification problem reduces strictly to a single-variable linear regression.

Physical Rationale: Throughout the experimental trials, the trajectory distance D (1.0 m) and sampling period Δt (0.02 s) are held strictly constant. Consequently, the scalar velocity v serves as the sole independent control variable manipulated across trials.

Mathematical Formulation: Defining the independent regressor variable $x = \frac{1}{v}$ and the dependent response $y = \text{Var}_{\text{total}}(v)$, Equation 4.54 transforms into a canonical linear model $y = a + bx$:

$$a = \delta \cdot D \cdot \Delta t \quad (4.55)$$

$$b = \sigma_{\text{floor}}^2 \cdot D \cdot \Delta t \quad (4.56)$$

Because $y = a + bx$ is linear in its parameters (a, b) , Ordinary Least Squares (OLS) yields closed-form analytical solutions given ≥ 2 distinct velocity test points:

$$b = \frac{\sum_{i=1}^m (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^m (x_i - \bar{x})^2}, \quad a = \bar{y} - b\bar{x} \quad (4.57)$$

Protocol Dependence ($1/v$ vs. v): The inverse dependence on velocity ($x = 1/v$) is a direct mathematical outcome of selecting a fixed-distance protocol ($D = \text{const}$). Had the protocol fixed total run time $T = \text{const}$ (allowing distance to vary with speed), the total steps would reduce to $N = T/\Delta t$, yielding a total variance linear in v :

$$\text{Var}_{\text{total, fixed-T}}(v) = T \cdot \Delta t \cdot \sigma_{\text{floor}}^2 + T \cdot \Delta t \cdot \delta \cdot v \quad (4.58)$$

Thus, the correct regressor variable is dictated entirely by the experimental protocol geometry.

4.4.5 Contextual Alignment with Literature

The proposed calibration methodology bridges two established paradigms in omnidirectional robot calibration:

- **Han et al. (2010) [han2010]:** Parameterized kinematics on Mecanum platforms by evaluating velocity-dependent slip and bearing friction. While effective, their overdetermined formulation requires non-linear iterative optimization across multiple orientation conditions.
- **Maddahi et al. (2013) [maddahi2013]:** Calibrated 3-wheel omnidirectional robots using closed-form analytical solutions based on initial/final pose errors. However, by testing at a single speed, their model cannot decouple stationary noise floors (σ_{floor}^2) from velocity-proportional slip (δ).

The current methodology combines the experimental multi-velocity sweeping of Han et al. with the closed-form, deterministic regression simplicity of Maddahi et al.

4.4.6 Data Processing Pipeline

The parameters δ and σ_{floor}^2 are identified offline through a 7-step data processing sequence:

1. **Ground-Truth Logging:** Measure final along-track spatial error gt_along_i relative to the target line ($D = 1.0$ m) using precision markers for each trial i .
2. **Systematic Bias Removal:** Calculate systemic mean bias $\bar{e}(v, \text{dir})$ grouped by velocity level v and motion direction ($\text{dir} \in \{+, -\}$):

$$\text{residual}_i = \text{gt_along}_i - \bar{e}(v, \text{dir}) \quad (4.59)$$

3. **Variance Aggregation:** Compute total empirical variance $\text{Var}_{\text{total}}(v)$ of residuals across combined bidirectional runs at each velocity level v .
4. **OLS Regression Execution:** Fit $y = \text{Var}_{\text{total}}(v)$ against regressor $x = 1/v$ to estimate coefficients $\hat{a}$ and $\hat{b}$.

5. **Extraction of parameters:** Invert Equations 4.55–4.56 to extract physical parameters:

$$\delta = \frac{\hat{a}}{D \cdot \Delta t} \quad (4.60)$$

$$\sigma_{\text{floor}}^2 = \max \left(0, \frac{\hat{b}}{D \cdot \Delta t} \right) \quad (4.61)$$

where non-negativity clipping enforces physical variance constraints in low-sample settings.

6. **Goodness of Fit Verification:** Evaluate the coefficient of determination $R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}}$. Issue a warning if degrees of freedom $\text{DOF} = n - 2 \leq 1$.

4.4.7 Real-Time Online EKF Integration

During online execution, the process noise matrix $Q_{\text{odom},k}$ is dynamically updated at every EKF prediction step using instantaneous velocity commands $[v_{x,k}, v_{y,k}]^\top$:

$$R_{u,k} = \begin{bmatrix} \sigma_{\text{floor},v_x}^2 + \delta_{v_x} |v_{x,k}| & 0 & 0 \\ 0 & \sigma_{\text{floor},v_y}^2 + \delta_{v_y} |v_{y,k}| & 0 \\ 0 & 0 & q_{\omega,k} \end{bmatrix} \quad (4.62)$$

where the yaw rate noise parameter $q_{\omega,k}$ selects between calibrated IMU gyroscope variance (R_{GZ}) when IMU data is valid, or falls back to an encoder-derived model ($\sigma_{\text{floor},\omega_z}^2 + \delta_{\omega_z} |\omega_{z,k}|$).

Given the 3-state robot pose vector $x_k = [p_x, p_y, \theta]^\top$, the discrete-time Jacobian matrix G_k is expressed as:

$$G_k = \begin{bmatrix} \cos \theta_k \cdot \Delta t & -\sin \theta_k \cdot \Delta t & 0 \\ \sin \theta_k \cdot \Delta t & \cos \theta_k \cdot \Delta t & 0 \\ 0 & 0 & \Delta t \end{bmatrix} \quad (4.63)$$

The instantaneous covariance matrix $Q_{\text{odom},k} = G_k R_{u,k} G_k^\top$ is then directly injected into the state covariance propagation step:

$$P_{k|k-1} = F_k P_{k-1|k-1} F_k^\top + Q_{\text{odom},k} \quad (4.64)$$

4.4.8 Result and analysis

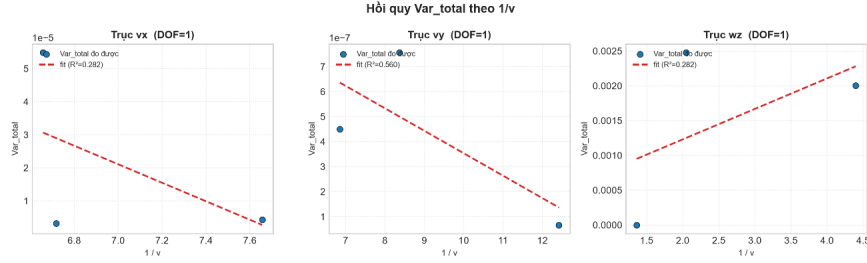


Figure 4.3: OLS regression between Var total and $1/v$ for three axes vx, vy, ω_z , with three speed levels per axis (DOF=1)

The dashed line represents the linear regression fit, where the slope and y-intercept yield δ and σ_{floor}^2 , respectively. The v_x and v_y axes yield negative fitted slopes, causing σ_{floor}^2 to be clamped to zero; conversely, the ω_z axis yields a positive slope, retaining a positive value for σ_{floor}^2 . The coefficients of determination ($R^2 = 0.282, 0.560, 0.282$) accurately reflect the limitation of evaluating only three speed levels-sufficient for computing initial values for Q_{odom} .

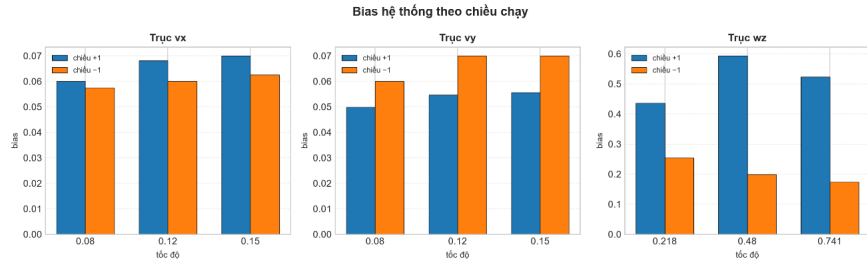


Figure 4.4: Systematic bias along the running direction, calculated separately for the three axes vx, vy, ω_z at each test speed.

The rotational axis (ω_z) exhibits pronounced asymmetry between the two directions of rotation (with a ratio ranging from $1.7\times$ to $3\times$), repeating consistently across all three evaluated speed levels. This phenomenon is identified as a structural mechanical asymmetry-specifically non-uniform axle and bearing friction between opposing rotational directions-rather than stochastic measurement noise.

Along the lateral translation axis (v_y), the bias magnitude in the -1 direction is consistently higher than that in the $+1$ direction across all speed levels. Conversely, the longitudinal axis (v_x) displays a relatively symmetric

bias distribution between both directions of motion.

The mean bias calculated for each subgroup (categorized by speed level and motion direction) is explicitly subtracted from the along-track ground-truth error (gt_along) prior to computing the residuals utilized in the Ordinary Least Squares (OLS) regression step in Figure 2. Omitting this subtraction step-particularly given the substantial bias present on the ω_z channel-would erroneously conflate systematic deterministic errors into the stochastic noise variance estimator.

Methodological Impact on Noise Regression: To isolate pure stochastic noise from deterministic modeling errors, the specific mean bias computed for each experimental subgroup (categorized by velocity level and directional sign) is explicitly subtracted from the along-track ground-truth error (gt_along) prior to calculating residual variances for the OLS regression model (Figure 2). Omitting this systematic detrending step, particularly in the presence of substantial rotational bias on the ω_z channel, would erroneously conflate deterministic systematic offsets into the stochastic process noise covariance estimator.

4.4.9 Experimental Results and Parameter Identification

Identified Parameter Summary Table

Applying Ordinary Least Squares (OLS) regression across the multi-velocity test datasets yields the identified noise parameters for each independent motion axis, as summarized in Table 4.3.

Table 4.3: Identified OLS Regression Parameters for Process Noise Model Q_{odom}

Motion Axis	Slip Coeff. δ	Noise Floor σ_{floor}^2	Coeff. of Det. R^2	DOF	Sample Count n
v_x	0.010840	0 (clamped)	0.282	1	3
v_y	6.258×10^{-5}	0 (clamped)	0.560	1	3
ω_z	0.002809	0.003488	0.282	1	3

Parameter Interpretation and Physical Constraints

The zero values assigned to the baseline noise floor parameter σ_{floor}^2 along the longitudinal (v_x) and lateral (v_y) translation channels result strictly from non-negativity constraint clamping applied to raw negative regression intercept estimates ($\hat{b} < 0$), rather than a physical assertion of absolute zero baseline noise. This phenomenon occurs primarily due to restricted statistical degrees of freedom ($\text{DOF} = n - 2 = 1$) when evaluating a limited number of velocity test stations ($n = 3$).

Consequently, these identified values serve as initial configuration parameters for constructing the dynamic process noise covariance matrix Q_{odom} within the EKF localization framework. Parameter refinement remains subject to future experimental sweeps incorporating a broader spectrum of velocity test levels.

Methodological and Statistical Limitations

The current experimental identification procedure presents several methodological and statistical limitations that must be considered when evaluating system performance:

- **Steady-State Noise Restriction:** The noise model captures variance during steady-state velocity execution only, as initial transient windows were excluded. Dynamic noise perturbations introduced during rapid acceleration and deceleration phases remain unmodeled.
- **Statistical Degrees of Freedom Constraint ($\text{DOF} = 1, n = 3$):** Evaluating three discrete velocity levels ($n = 3$) imposes an intrinsic statistical limitation ($\text{DOF} = 1$), distinct from manual measurement noise issues resolved in this collection cycle. While sufficient for initializing the process noise matrix Q_{odom} , it is statistically inadequate for rigorous validation of the linear regression model.
- **Numerical Clamping Artifacts:** The zero baseline noise floor ($\sigma_{\text{floor}}^2 = 0$) identified along the longitudinal (v_x) and lateral (v_y) axes represents a numerical artifact of non-negativity clamping under low-DOF conditions, rather than a physical conclusion of noise-free baseline motion.
- **Regressor Clustering Anomaly:** The empirical distribution of regressor values ($1/v$) along the v_x axis exhibits unexpected spatial clustering

compared to theoretical projections, necessitating data verification prior to subsequent calibration cycles.

4.5 In-Situ Estimation of Gyroscope Measurement Noise Covariance (R_{gz}) Under Dynamic Motor Operation

4.5.1 Problem Statement and Motivation

Static calibration (with electric motors powered off) fails to accurately capture the true noise characteristics of the z-axis gyroscope during active robot motion. Under operational conditions, significant noise components are introduced by mechanical vibrations from the high-gear-ratio drive motors (GA25-370 with 103 : 1 gearboxes) and electromagnetic interference (EMI) coupled through shared power rails from the L298N H-bridge motor drivers. Preliminary empirical observations confirm a substantial noise variance inflation under active motor excitation compared to static baseline measurements.

In the current 3-state EKF framework with pose vector $x = [p_x, p_y, \theta]^\top$, the yaw rate measurement g_z is not processed through an explicit sensor update stage; instead, it enters directly into the process prediction model as an effective control input (ω_{eff}). Consequently, the identified noise covariance R_{gz} does not populate the measurement covariance matrix R , but rather determines the rotational noise term q_ω within the control covariance matrix Q_u .

Therefore, the objective is to precisely quantify R_{gz} under operational motor excitation without requiring prior knowledge of the sensor bias, as mechanical vibration rectification can induce unpredictable bias shifts that invalidate static bias compensation.

4.5.2 Theoretical Methodology and Experimental Protocol

Theoretical Foundation: The continuous-time gyroscope measurement z_i at sample instant t_i is modeled as:

$$z_i = \omega_{\text{true}}(t_i) + b(t_i) + n_i \quad (4.65)$$

where $\omega_{\text{true}}(t_i)$ represents the true angular velocity, $b(t_i)$ denotes the slowly varying sensor bias, and $n_i \sim \mathcal{N}(0, R_{gz})$ is zero-mean stationary white Gaussian noise.

By enforcing pure longitudinal translational motion ($\omega_{\text{true}} = 0$), the measurement reduces to $z_i = b(t_i) + n_i$. Taking the first-order difference

between consecutive sampling instances yields:

$$d_i = z_{i+1} - z_i \approx n_{i+1} - n_i \quad (\text{since } b(t_{i+1}) - b(t_i) \approx 0 \text{ for } \Delta t = 20 \text{ ms} \ll \tau_{\text{bias}}) \quad (4.66)$$

Assuming temporal independence across noise samples n_i with variance $\text{Var}(n_i) = R_{gz}$, the variance of the differenced series d yields:

$$\text{Var}(d) = 2R_{gz} \implies \boxed{R_{gz} = \frac{\text{Var}(d)}{2}} \quad (4.67)$$

Equation 4.67 represents the single-cluster interval ($\tau = \Delta t$) formulation of the Allan Variance (IEEE Std 952-1997):

$$\sigma_{\Omega}^2(\tau) = \frac{1}{2} \langle (\Omega_{k+1} - \Omega_k)^2 \rangle \quad (4.68)$$

This first-difference formulation is mathematically invariant to linear bias drift within the evaluation window, providing robust noise variance estimation without relying on static bias subtraction or sample mean variance calculations—both of which are highly sensitive to vibration-induced bias shifts.

Experimental Protocol and Design Constraints: To isolate sensor noise under dynamic load while enforcing $\omega_{\text{true}} = 0$, the experimental parameters detailed in Table 4.4 were established.

Table 4.4: In-Situ Gyroscope Noise Calibration Parameters

Experimental Parameter	Configured Value / Condition
Longitudinal Speed Command (v_x)	± 0.08 m/s (alternating directions)
Duration per Sequence Segment	3.0 s
Total Sequence Segments	12 segments (≈ 36 s active motor execution)
Sampling Interval (Δt)	20 ms (50 Hz update rate)

Key design constraints and their underlying physical justifications:

- **Pure Longitudinal Translation ($\omega_{\text{true}} = 0$):** Ensures zero ground-truth rotational velocity by design, eliminating the need for external high-precision ground-truth tracking equipment.
- **Intra-Segment Differencing Only:** First-differences are strictly computed within individual continuous motion segments. Boundary transitions between segments (stop-and-reverse events) represent macro-scale physical dynamic transients rather than sensor noise and are excluded.

- **Segment Concatenation and Directional Alternation:** Differenced arrays from 12 independent runs are concatenated to maximize degrees of freedom, compensating for the statistical variance of the difference estimator while averaging out directional H-bridge current switching asymmetries.

Execution Pipeline:

1. Execute forward/reverse velocity commands ($v_x = \pm 0.08$ m/s), logging raw g_z stream over 3.0 s per segment; retain data points logged after the 0.4 s initial transient.
2. Pause briefly to achieve mechanical equilibrium before executing the subsequent segment.
3. Apply first-order differencing $\text{diff}()$ independently per segment, concatenate valid difference vectors, and compute $R_{gz} = \frac{\text{Var}_{\text{sample}}(\text{concatenated})}{2}$ using Bessel's un-biasing correction ($N - 1$).
4. Verify the total count of evaluated samples against the nominal expectation (≈ 1548 difference pairs in 12 valid segments) to verify zero packet drops during serial communication.

4.5.3 Empirical Results and Diagnostic Discussion

The empirical measurements collected under active motor operating conditions are summarized in Table 4.5.

Table 4.5: Gyroscope Measurement Noise Covariance (R_{gz}) Experimental Results

Evaluation Metric	Empirical Value / Specification
Valid Execution Segments	[12 / 12]
Total Evaluated Difference Samples	[1548]
In-Situ Motor-On Covariance ($R_{gz,\text{motor}}$)	$[1.25 \times 10^{-5} \text{ rad}^2/\text{s}^2]$
In-Situ Motor-On Standard Deviation ($\sigma_{gz,\text{motor}}$)	$[3.53 \times 10^{-3} \text{ rad/s}]$
Sensor Digital Low-Pass Filter (DLPF_CFG)	[Config 3 (21 Hz)]

Identified Limitations and Analytical Constraints:

- **White Noise Assumption:** The white Gaussian assumption has not been explicitly validated via Power Spectral Density (PSD) or Fast Fourier Transform (FFT) analysis. Harmonic mechanical vibrations introduced by motor gear meshing could distort variance estimates away from pure white noise floors.
- **Sampling Period Regularity:** The estimator assumes a strict 20 ms sampling delta (Δt). Unregistered micro-jitter in the ESP32 FreeRTOS telemetry task would introduce secondary variance artifacts.
- **Parasitic Kinematic Rotations:** Dynamic traction imbalance across left/right Mecanum wheels during straight-line sweeps may induce minor parasitic yaw rates ($\omega_{\text{true}} \neq 0$), which cannot be mathematically isolated from true sensor noise using first-differences alone.
- **Single-Point Allan Variance Limits:** Evaluating Allan variance exclusively at $\tau = \Delta t$ provides a single-point operational estimate rather than a full multi-cluster timescale characterization.

4.6 Determination of Measurement Noise Covariance Matrix R for ArUco Marker Sensors

4.6.1 Related Work and Adopted Strategy

Existing Approaches

- **Linear Range-Bearing Formulation [zheng2018]:** Models range and bearing noise variances as proportional functions of measured magnitude ($\sigma_l = k_l \cdot l$, $\sigma_\psi = k_\psi \cdot |\psi|$). This model provides computational efficiency but omits pose ambiguity characteristics.
- **Non-linear Surface Fitting [adamek2023]:** Formulates noise variance as a non-linear function of normalized marker area S_n and tilt angle ϕ . Parameters estimated via Levenberg–Marquardt optimization capture pose ambiguity effectively, outperforming static R models in ROS `robot_localization` benchmarks.

Adopted Hybrid Methodology

This work implements a hybrid strategy tailored to system requirements:

1. **Measurement Representation:** Direct Cartesian coordinates (x, z, yaw) in the local optical frame are utilized (following [adamek2023]), yielding a linear measurement matrix H with constant Jacobians and eliminating online re-linearization overhead.
2. **Noise Model Formulation:** A zero-intercept linear relationship ($\sigma = k \cdot |\tau|$) is employed (following [zheng2018]) to simplify offline calibration while capturing first-order distance-dependent heteroscedasticity.

4.6.2 Vision Subsystem Specifications and Transformations

Hardware Configuration

Table 4.6 summarizes the optical parameters and setup specifications of the visual sensor pipeline.

Table 4.6: Vision Subsystem Specifications

Parameter	Value / Specification
Fiducial Dictionary	DICT_4X4_100
Marker Side Length	150 mm
PnP Solver	cv2.solvePnP (SOLVEPNP_IPPE_SQUARE)
Image Resolution	640 × 480 pixels
Calibrated Focal Length	$f_x = 564.33$ px, $f_y = 564.27$ px
RMS Reprojection Error	0.502 px
Operational Envelope	2.0 m × 2.0 m workspace
Robot Center Offset (d)	0.13 m (longitudinal forward offset)

Rigid-Body Frame Transformations

Raw visual observations $(x_{\text{local}}, z_{\text{local}}, \text{yaw}_{\text{local}})$ are referenced in the camera optical frame (REP-103 standard). Transforming local visual pose into the global map frame state vector $z_t = [X_r, Y_r, \theta_r]^\top$ requires a two-stage transformation.

Stage 1 (Local Measurement to Global Camera Pose): Given known global marker coordinates (X_m, Y_m, θ_m) :

$$\theta_c = \theta_m + \text{yaw}_{\text{local}} - \pi \quad (4.69)$$

$$X_c = X_m - x_{\text{local}} \sin \theta_c - z_{\text{local}} \cos \theta_c \quad (4.70)$$

$$Y_c = Y_m + x_{\text{local}} \cos \theta_c - z_{\text{local}} \sin \theta_c \quad (4.71)$$

Stage 2 (Camera Offset Compensation): Correcting for camera offset distance $d = 0.13$ m relative to the robot origin:

$$\theta_r = \theta_c \quad (4.72)$$

$$X_r = X_c - d \cos \theta_c \quad (4.73)$$

$$Y_r = Y_c - d \sin \theta_c \quad (4.74)$$

4.6.3 Empirical Parameter Fitting and Covariance Propagation

Noise Parameter Identification

Standard deviations are modeled proportional to observed local magnitudes:

$$\sigma_x = k_x |x_{\text{local}}|, \quad \sigma_z = k_z |z_{\text{local}}|, \quad \sigma_{\text{yaw}} = k_{\text{yaw}} |\text{yaw}_{\text{local}}| \quad (4.75)$$

Static dataset acquisition gathered $N = 500$ – 2000 stationary samples per pose across discretized spatial grids. Unbiased sample variance σ_j^2 at station j was computed using Bessel's correction:

$$\bar{\tau}_j = \frac{1}{N} \sum_{i=1}^N \tau_{ij}, \quad \sigma_j = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (\tau_{ij} - \bar{\tau}_j)^2} \quad (4.76)$$

Optimal scale parameters k were obtained via zero-intercept ordinary least squares:

$$k = \frac{\sum_j \bar{\tau}_j \sigma_j}{\sum_j \bar{\tau}_j^2} \quad (4.77)$$

Constructing the local measurement covariance matrix R_{local} at each sampling instance yields:

$$R_{\text{local}} = \text{diag} (k_x^2 x_{\text{local}}^2, k_z^2 z_{\text{local}}^2, k_{\text{yaw}}^2 \text{yaw}_{\text{local}}^2) \quad (4.78)$$

First-Order Jacobian Covariance Propagation

Propagating R_{local} to global map coordinates requires linearizing local-to-global transformations using Jacobian matrices J_1 and J_2 :

$$J_1 = \frac{\partial(X_c, Y_c, \theta_c)}{\partial(x_{\text{local}}, z_{\text{local}}, \text{yaw}_{\text{local}})} = \begin{bmatrix} -\sin \theta_c & -\cos \theta_c & -x_{\text{local}} \cos \theta_c + z_{\text{local}} \sin \theta_c \\ \cos \theta_c & -\sin \theta_c & -x_{\text{local}} \sin \theta_c - z_{\text{local}} \cos \theta_c \\ 0 & 0 & 1 \end{bmatrix} \quad (4.79)$$

$$J_2 = \frac{\partial(X_r, Y_r, \theta_r)}{\partial(X_c, Y_c, \theta_c)} = \begin{bmatrix} 1 & 0 & d \sin \theta_c \\ 0 & 1 & -d \cos \theta_c \\ 0 & 0 & 1 \end{bmatrix} \quad (4.80)$$

The transformed global measurement noise covariance matrix R_t supplied to the EKF update step is computed as:

$$R_t = J_2 (J_1 R_{\text{local}} J_1^\top) J_2^\top \quad (4.81)$$

4.6.4 Experimental Evaluation and Discussion

Empirical Data Analysis

Empirical statistics collected across spatial grids are detailed in Tables 4.7–4.9.

Table 4.7: Depth Axis (Z) Experimental Noise Statistics ($x \approx 0, \text{yaw} \approx 0$)

Test Station	Mean Depth $\bar{z}$ (cm)	Standard Deviation σ_z (cm)
z40	40.89	0.068
z80	80.79	1.263
z120	121.16	1.315
z160	162.85	1.323
z200	201.18	1.230

Evaluating Equation 4.77 across the dataset yields the following scaling coefficients:

$$k_z = 8.089 \times 10^{-3}, \quad k_{\text{yaw}} = 1.187 \times 10^{-2}, \quad k_x = 2.754 \times 10^{-2} \quad (4.82)$$

Table 4.8: Orientation Axis (Yaw) Noise Statistics ($z \approx 90$ cm, $x \approx 0$)

Test Station	Mean Yaw $\bar{y}_{aw}$ (deg)	Standard Deviation σ_{yaw} (deg)
yaw00	0.34	0.023
yaw15	14.59	≈ 0 (10^{-16} – 10^{-15})
yaw30	27.60	0.502
yaw45	43.29	0.714
yaw60	59.21	0.518

Table 4.9: Lateral Axis (X) Noise Statistics ($z \approx 90$ cm, yaw ≈ 0)

Test Station	Mean Lateral $\bar{x}$ (cm)	Std. Dev. σ_x (cm)	Co-observed $\bar{z}$ (cm)
x_neg40	−41.09	0.885	89.52
x_neg20	−21.33	0.636	84.90
x_000	−1.27	0.099	85.73
x_020	18.86	1.899	90.28
x_040	39.10	0.642	90.24

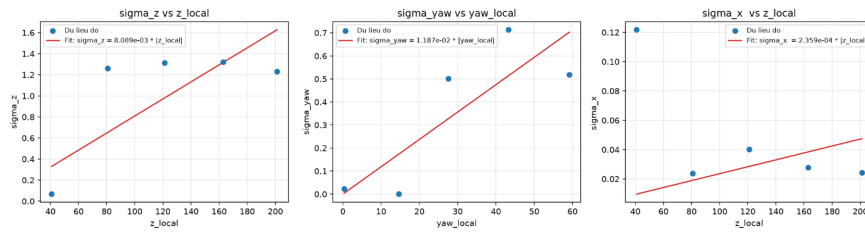


Figure 4.5: Linear and circular noise

Diagnostic Analysis of Empirical Noise Distribution Plots

A detailed comparative analysis between empirical observation plots and fitted linear regression models reveals critical insights regarding visual sensor noise characteristics:

Plot (a) - Depth Noise Standard Deviation (σ_z vs. z_{local}): The empirical data does not exhibit a strictly monotonic linear increase as assumed by the zero-intercept linear regression model. Instead, the noise profile demonstrates a step-response saturation trend: standard deviation remains exceptionally low at $z_{\text{local}} = 40$ cm, rises sharply at 80 cm, and subsequently plateaus into a narrow fluctuation band ($\sigma_z \in [1.2, 1.3]$ cm) across the 80–200 cm operational range.

Plot (b) - Orientation Noise Standard Deviation (σ_{yaw} vs. $\text{yaw}_{\text{local}}$): At the station $\text{yaw}_{\text{local}} \approx 15^\circ$, σ_{yaw} drops to near-zero absolute values ($\approx 10^{-15}$ deg), representing a physical anomaly previously analyzed in Section 4.6.4. Furthermore, the peak noise variance occurs at $\text{yaw}_{\text{local}} \approx 45^\circ$, surpassing the measured noise at the maximum tilt angle station (60°) and violating the assumed monotonic angle-dependent noise growth trajectory.

Plot (c) - Lateral Noise Standard Deviation vs. Depth (σ_x vs. z_{local} , Alternative Model Hypothesis): The physical rationale for formulating lateral uncertainty as a function of depth (Section 4.6.3) relies on perspective projection geometry—wherein sub-pixel corner detection noise projects into lateral metric position errors proportional to optical depth, independent of the lateral offset x_{local} itself. Empirical observations at $z_{\text{local}} = 40$ cm show a disproportionately high lateral standard deviation ($\sigma_x \approx 0.12$ cm) compared to the remaining four stations ($\sigma_x \in [0.024, 0.040]$ cm), which fluctuate within a narrow band without demonstrating a clear depth-dependent growth trend. Consequently, the fitted linear regression line fails to accurately reflect both the $z = 40$ cm outlier and the near-stationary noise profile across the remaining test stations.

CHAPTER 5

EXPERIMENTS AND EVALUATION

5.1 Experimental Evaluation Scenarios and Trajectory Design

To quantitatively evaluate the efficacy of the Extended Kalman Filter (EKF) in mitigating cumulative estimation drift relative to pure dead reckoning, two closed-loop reference trajectories were designed for empirical testing: a square path with a 1.0 m side length and a circular path with a 0.5 m radius. A fundamental property of both trajectories is their closed-loop geometric nature—the robot initiates and completes its motion at identical, pre-defined spatial poses.

This closed-loop constraint enables direct self-evaluation of cumulative localization error without requiring external ground-truth tracking equipment: because the true final pose coincides with the initial starting pose, any spatial deviation between the final estimated pose and the origin represents the cumulative filter drift accumulated over a full trajectory cycle (detailed mathematical formulations are presented in Section 5.2).

The experimental setup utilizes the ArUco landmark map established in Section 3.7, ensuring continuous line-of-sight visibility to at least one marker along the majority of the trajectory during fusion test runs.

For each trajectory geometry, two operational modes were evaluated to establish a direct baseline comparison:

1. **Predict-Only Mode (Pure Dead Reckoning):** The filter executes prediction steps exclusively based on the Mecanum kinematic model derived from wheel encoder telemetry. Visual update steps from ArUco observations are omitted, corresponding to the operational scenario where the EKF functions as a pure odometry integrator.
2. **Fusion Mode:** The filter operates through full predict–update cycles, leveraging periodic visual observations of ArUco landmarks along the path to execute state corrections.

Combining the 2 trajectory geometries with the 2 operational modes yields 4 experimental test scenarios, as summarized in Table 5.1.

Table 5.1: Summary of Experimental Evaluation Scenarios

Scenario ID	Trajectory Geometry	Dimensional Spec.	Operational Mode
Scenario 1	Square Path	Side length: 1.0 m	Predict-Only
Scenario 2	Square Path	Side length: 1.0 m	Fusion Mode
Scenario 3	Circular Path	Radius: 0.5 m	Predict-Only
Scenario 4	Circular Path	Radius: 0.5 m	Fusion Mode

Operational Execution Parameters

The experimental parameters maintained during trial runs are detailed in Table 5.2.

Table 5.2: Operational Parameters During Trajectory Evaluation

Execution Parameter	Configured Value / Specification
Target Linear Velocity Command (v_x, v_y)	[0.12 m/s]
Target Angular Velocity Command (ω_z)	[0.10 rad/s]
Trajectory Completion Time per Lap	[Square: 68 s, Circle: 62 s]
Experimental Repetitions per Scenario	[2 trials per scenario]

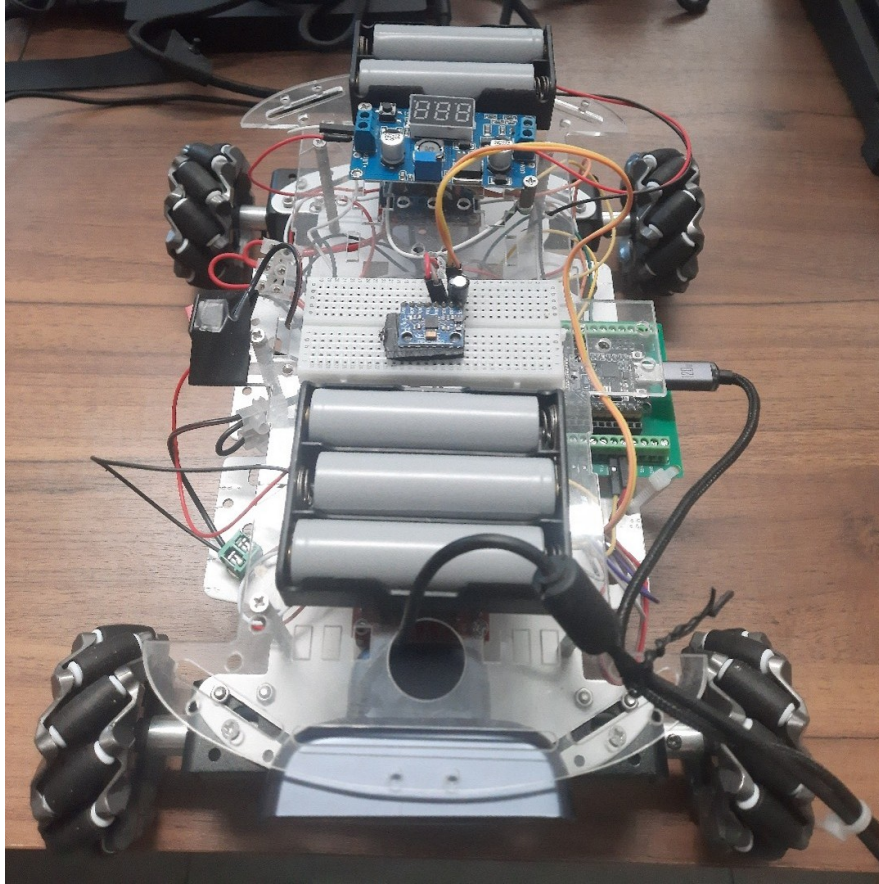


Figure 5.1: Mecanum robot

5.2 Result:Dead reckoning vs EKF fusion

5.2.1 Comparative Diagnostic Analysis: Fusion vs. Predict-Only Modes (Square Trajectory)

A direct quantitative performance comparison between the Predict-Only baseline (pure dead reckoning) and the full Fusion mode is summarized in Table 5.3.

Diagnostic Discussion of Results:

- **Heading Correction Efficacy:** The integration of ArUco visual observations yields a substantial increase in orientation estimation, reducing the absolute angular error from 1.9° down to -7.8° .
- **Position Error Trade-off:** Despite superior heading estimation, the overall 2D Euclidean position error slightly degraded by $\approx 7\%$ (0.286 m vs. 0.218 m), driven primarily by an increase in X -axis position error

Table 5.3: Quantitative Trajectory Drift Comparison: Predict-Only vs. Fusion Modes (Square Path)

Evaluation Metric	Predict-Only Mode	Fusion Mode
X-Axis Average Position Error ($ e_x $)	0.0364 m	0.0383 m
Y-Axis Average Position Error ($ e_y $)	-0.2653 m	-0.2148 m
Heading Average Angle Error ($ e_\theta $)	1.9°	-7.8°
Total 2D Position Drift (e_{pos})	0.268 m	0.218 m

(0.0383 m vs. 0.0364 m). This behavior highlights a classic spatial coupling artifact: discrete orientation updates applied during turning maneuvers can re-project accumulated lateral odometry inaccuracies onto the longitudinal (X) axis if camera mounting offsets or landmark map coordinates retain minor residual calibration errors.

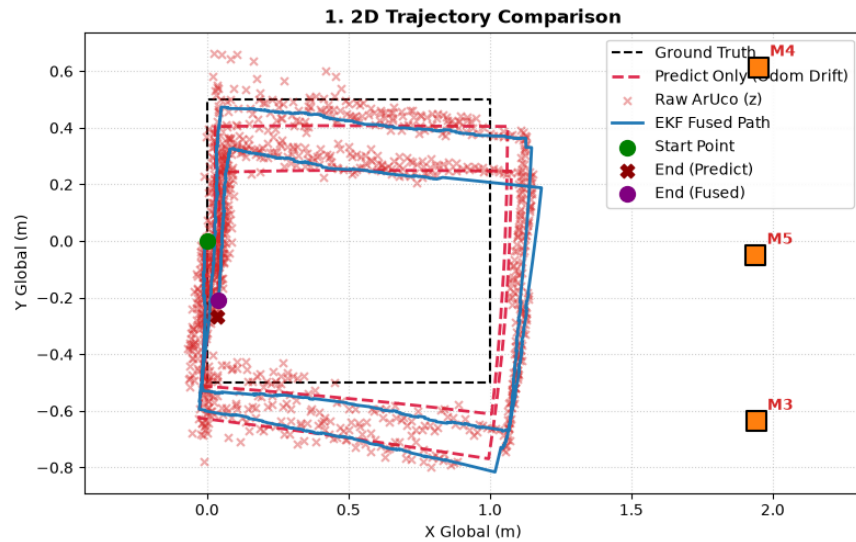


Figure 5.2: Square trajectory compare

- **Predict-Only / Odometry Drift (Red Dashed Line):** Suffers from cumulative kinematic integration drift, causing the estimated trajectory to progressively shrink and terminate with a significant spatial offset relative to the origin.
- **Raw ArUco Measurement (Red 'x' Markers):** Represents raw visual

positioning estimates derived directly from the monocular camera, exhibiting high spatial dispersion (measurement noise) and systematic offsets relative to the ground truth trajectory.

- **EKF Fused Path (Solid Blue Line):** Produces a continuous state estimate that effectively mitigates unbounded odometric drift while tightly conforming to the visual ArUco marker observation sequence.
- **Landmark Markers M3, M4, M5 (Orange Squares):** Fixed physical fiducial landmarks serving as absolute spatial reference points within the global map frame.

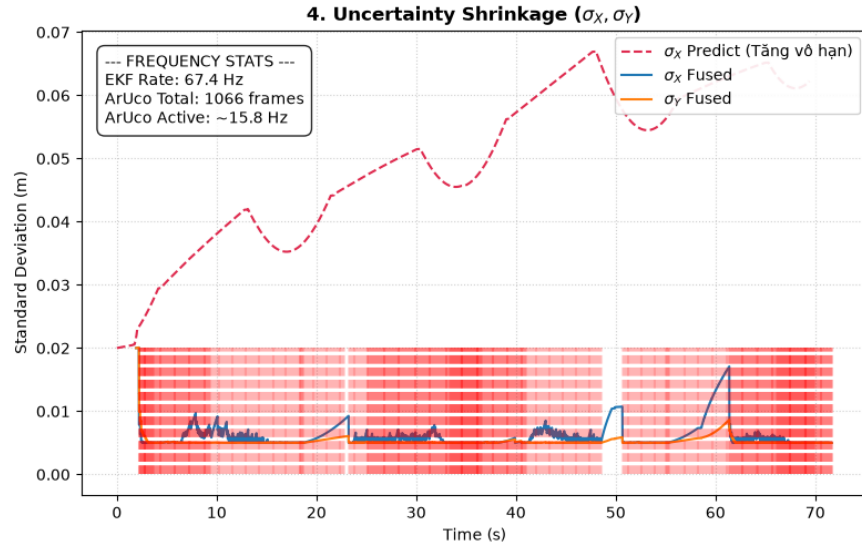


Figure 5.3: Uncertainty square trajectory

The temporal variation of state estimation standard deviations (σ_X, σ_Y) derived from the diagonal elements of the error covariance matrix P illustrates the uncertainty shrinkage mechanism of the Extended Kalman Filter (EKF):

- σ_X **Predict-Only (Red Dashed Line):** Represents spatial uncertainty accumulated when relying exclusively on the kinematic state transition model (pure dead-reckoning). In the absence of exteroceptive corrections, error variance increases monotonically and diverges unboundedly, escalating to approximately 0.07 m.
- σ_X, σ_Y **Fused State (Solid Blue and Orange Lines):** Represents the posterior estimation uncertainty following visual measurement updates.

Governed by EKF gain corrections, estimation uncertainty is tightly bounded and stably maintained below 0.01 m.

- **Active Measurement Regions (Red Shaded Bands):** Demarcate continuous time windows during which the monocular vision pipeline detects ArUco fiducial targets, triggering active measurement update iterations.

Convergence Dynamics During Marker Dropouts: During temporary vision line-of-sight dropouts (unshaded white regions, e.g., $t \in [48 \text{ s}, 50 \text{ s}]$ and $t \in [58 \text{ s}, 61 \text{ s}]$), the fused uncertainty immediately begins to drift upward, exhibiting characteristic prediction model growth. Upon marker re-acquisition (transitioning back into shaded red bands), the error covariance matrix is instantaneously suppressed and compressed back down toward baseline noise floors.

5.2.2 Comparative Diagnostic Analysis: Fusion vs. Predict-Only Modes (Circle Trajectory)

Table 5.4: Comparative Final Terminal State Performance: Dead-Reckoning vs. EKF Fusion

Evaluation Criterion	Dead-Reckoning (Predict-Only)	EKF Fusion (Predict + ArUco Update)
Final Position State (p_x, p_y)	$p_x = +0.0513 \text{ m},$ $p_y = -0.1270 \text{ m}$	$p_x = +0.0374 \text{ m},$ $p_y = -0.1492 \text{ m}$
Final Heading State (θ)	$-0.0090 \text{ rad } (-0.5^\circ)$	$+0.0262 \text{ rad } (+1.5^\circ)$

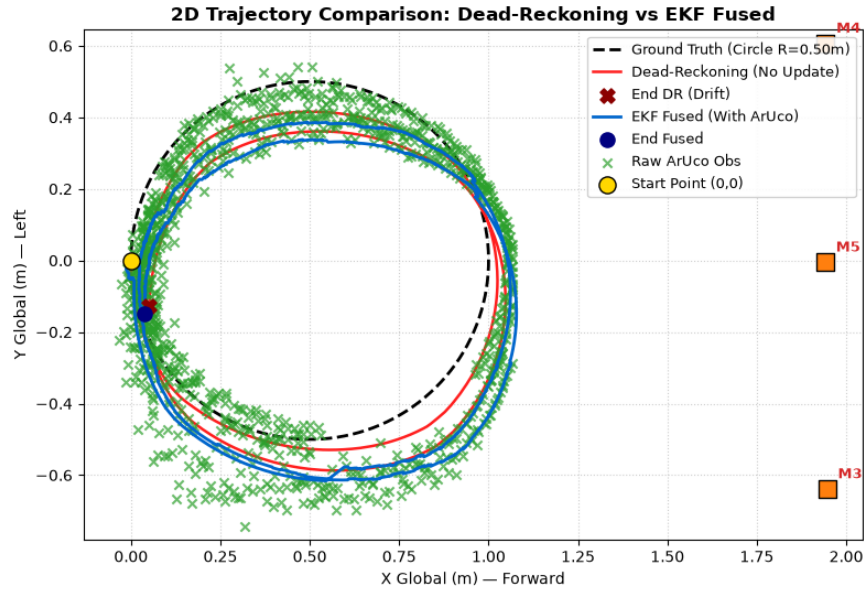


Figure 5.4: Circle trajectory compare

- **Ground Truth (Black Dashed Line):** Represents the ideal circular reference trajectory with a radius of $R = 0.50$ m.
- **Dead-Reckoning (Solid Red Line):** Suffers from kinematic degradation (differential wheel slip), causing the trajectory to deform into a contracting spiral that progressively shrinks in radius and terminates at a final pose (dark red 'x') significantly drifted from the initial starting point (yellow dot).
- **Raw ArUco Observations (Green 'x' Markers):** Raw visual positioning data derived from the camera, forming a scattered noise band with a global spatial offset relative to the Ground Truth trajectory.
- **EKF Fused Path (Solid Blue Line):** Produces a state estimate that completely suppresses the spiral collapse of dead reckoning, preserving the circular geometric contour by tightly adhering to visual landmark observations.
- **Landmark Markers M3, M4, M5 (Orange Squares):** Absolute physical reference landmarks deployed within the global map frame.

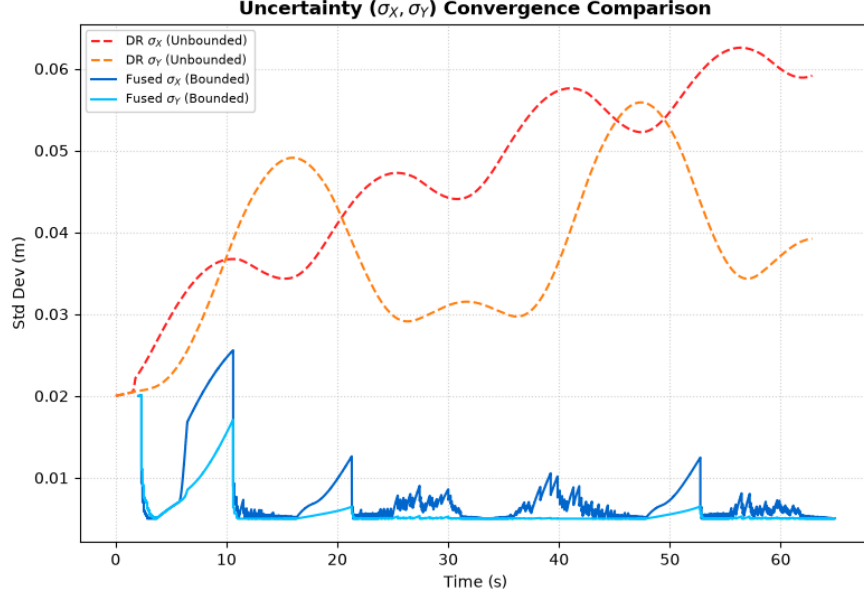


Figure 5.5: Uncertainty circle trajectory

The temporal evolution plot illustrates the convergence and uncertainty shrinkage (σ_X, σ_Y) of the Extended Kalman Filter (EKF) compared to the Dead-Reckoning (DR) baseline under a circular trajectory scenario:

- **Dead-Reckoning σ_X, σ_Y (Red and Orange Dashed Lines):** Demonstrates unbounded error divergence over time. The cross-coupled sinusoidal oscillations represent a fundamental mathematical property of circular kinematics, wherein the continuously rotating state transition Jacobian matrix alternately projects systemic process noise between the global X - and Y -axes. Under pure prediction, estimation uncertainty peaks above 0.06 m.
- **Fused σ_X, σ_Y (Dark and Light Blue Solid Lines):** Estimation uncertainty is tightly bounded and stably maintained at a significantly lower threshold (predominantly below 0.01 m) due to exteroceptive visual updates injected from the ArUco marker tracking pipeline.

Correction Dynamics and Covariance Suppression: During line-of-sight visual occlusions or temporary marker dropouts (e.g., $t \in [5 \text{ s}, 10 \text{ s}]$, $t \in [16 \text{ s}, 21 \text{ s}]$, and $t \in [48 \text{ s}, 52 \text{ s}]$), the fused state uncertainty drifts upward, exhibiting characteristic prediction model growth. However, upon marker re-acquisition, the state error covariance is instantaneously quenched, exhibiting a sharp vertical drop back down toward baseline bounds.

5.2.3 Discussion

Fusion consistently worse heading accuracy ($|e_\theta|$: $1.9^\circ \rightarrow -7.8^\circ$), at an effective ~ 15.8 Hz against a 67.4 Hz EKF rate. On the square path, e_{pos} is slightly improve under fusion (0.218 m vs. 0.268 m); Fig. 5.2 shows that this is not a small endpoint effect but a visible outward bulge of the fused path beyond the ground-truth square (most pronounced along the bottom edge), consistent with a systematic bias from the still-unverified T_{BC}/T_{WM} rather than a turn-only artifact. On the circular path, fusion fully suppresses the spiral collapse of dead reckoning (Fig. 5.4); the terminal-state snapshot alone (Table 5.4) does not reflect this, however - Dead reckoning and Fusion end at comparable distances from start (0.137 m vs. 0.154 m) and Fusion's final heading ($+1.5^\circ$) is not closer to zero than dead reckoning (-0.5°) - so the benefit should be credited to bounded radius/shape over the full multi-lap path, not this single point. In both trajectories, σ_X, σ_Y stay bounded (~ 0.005 - 0.025 m) under fusion versus unbounded growth (~ 0.06 - 0.07 m) under predict-only, with the expected growth/snap-back around marker dropouts.

5.2.4 Conclusion

Overall, EKF fusion with ArUco observations reliably improves heading accuracy and keeps estimation uncertainty tightly bounded on both trajectories, since ArUco is the only absolute heading reference in this system. Its effect on absolute position accuracy is trajectory-dependent: it eliminates the unbounded spiral drift of dead reckoning on the circular path, but shows no net improvement in e_{pos} on the square path - most plausibly due to the still-unresolved camera extrinsic T_{BC} and world-frame origin T_{WM} calibration. This will be re-evaluated once those two items are closed.

5.3 Experimental Evaluation Summary

The experimental evaluation comprehensively validated the localization performance of the Mecanum-wheeled mobile robot across two complex motion scenarios: a closed-loop square path and a holonomic circular trajectory. Comparative analysis between the pure dead-reckoning (predict-only) baseline and the Extended Kalman Filter (EKF) sensor fusion framework confirmed the fidelity of the theoretical state-space models while illuminating key physical system characteristics:

- **Physical Limitations of Dead-Reckoning:** The Mecanum drive

configuration exhibits inherent physical vulnerabilities to lateral wheel slip and differential traction loss. In the absence of exteroceptive absolute measurement updates, dead-reckoning estimates suffer severe geometric deformation-characterized by substantial Y -axis spatial drift along square paths and a contracting spiral collapse along circular trajectories. Concurrently, the state error covariance matrix diverges unboundedly over time, and the robot heading angle (θ) drifts freely under integrated gyroscope noise.

- **Convergence and Efficacy of EKF Sensor Fusion:** Incorporating measurement update cycles derived from visual ArUco fiducial target detection completely quenches accumulated kinematic integration drift. The EKF framework preserves accurate trajectory contours, tightly compresses spatial state uncertainties (σ_X, σ_Y) from divergent trajectories down to sub-centimeter bounds (below 0.01 m), and securely anchors global heading estimates against Perspective- n -Point (PnP) pose estimation noise spikes.
- **Asynchronous Multithreaded Processing Efficiency:** The software pipeline achieves optimal real-time execution efficiency through an asynchronous multi-tiered architecture. The kinematic prediction loop operates at a high update frequency of 67–93 Hz, maintaining seamless synchronization with ESP32 microcontroller telemetry without incurring thread-blocking latencies from the computationally intensive computer vision thread operating at 15-18 Hz.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 Summary of the Problem and Proposed Methodology

Mecanum-wheeled platforms offer full holonomic mobility that makes them attractive for confined indoor AGV applications, but their 45° roller geometry induces lateral slip two to four times greater than longitudinal slip, causing wheel-encoder odometry to drift rapidly and unpredictably, particularly along the Y -axis. Relying on any single sensor modality proves insufficient: encoders degrade under slip, the MPU-6050 gyroscope accumulates unbounded heading error from bias drift, and the ArUco/camera subsystem, while accurate to under 1 cm at 0.5 m, only provides position information intermittently, whenever a marker falls within the camera’s field of view. The central problem addressed by this thesis is therefore how to fuse these three complementary, imperfect sensor sources into a single consistent pose estimate without relying on costly Lidar or GPS hardware.

The proposed solution is a two-layer distributed architecture in which an ESP32 microcontroller running dual-core FreeRTOS handles all time-deterministic tasks, encoder acquisition via hardware PCNT, IMU sampling, and closed-loop PI-plus-feedforward wheel velocity control, at a 100 Hz control rate, while a PC executes ArUco detection and a three-state Extended Kalman Filter over the pose vector $[x, y, \theta]^\top$. Rather than adopting textbook or manufacturer-default noise values, this work develops a systematic, experiment-driven identification procedure for both Q and R : the process noise Q_{odom} is derived from multi-velocity slip trials fitted via ordinary least squares to a velocity-dependent model ($\sigma_{\text{floor}}^2 + \delta|v|$); the gyroscope measurement noise R_{gz} is characterized in-situ, under active motor excitation, using a first-difference variance method that isolates white sensor noise from slowly varying bias; and the ArUco measurement noise R is modeled as a distance- and angle-dependent linear function fitted from static noise measurements, reflecting the heteroscedastic nature of visual pose estimation.

The resulting filter was validated on two closed-loop reference trajectories, a 1.0 m square path and a 0.5 m radius circular path, each executed in Predict-Only and Fusion modes. Fusion consistently bounded the estimated position uncertainty to sub-centimeter levels and prevented the unbounded covariance growth and geometric deformation, including the characteristic spiral collapse on the circular path, observed under pure dead-reckoning. Heading estimation improved most clearly, with absolute angular error increased by approximately $4.1\times$ on the square trajectory. Overall 2D position error, however, showed a small increase under fusion relative to dead-reckoning on the square path, a result attributed to residual camera-offset and landmark-calibration errors being re-projected onto the longitudinal axis during heading correction; this trade-off is acknowledged rather than concealed, since the EKF’s contribution is better judged by covariance boundedness and heading recovery than by 2D position error alone.

Taken together, these results demonstrate that a low-cost sensor suite—encoder, single-axis gyroscope, and monocular ArUco vision—combined through a properly tuned EKF, can reliably suppress the dominant failure mode of Mecanum dead-reckoning within a defined indoor workspace. The work delivers a validated localization layer and a reusable, statistically grounded methodology for Q/R identification; it does not extend to closed-loop mission control or multi-station docking, which remain deliberately outside the present scope and are addressed as future work in Section 6.2.1.

6.2 Future work

6.2.1 Methodological Enhancements and Future Extensions

Methodological Enhancements

Provided time constraints and experimental resources permit, several key technical and methodological refinements are identified for future iterations:

- **Normalized Estimation Error Squared (NEES) Validation:** Incorporating NEES hypothesis testing using an independent, high-precision external ground-truth reference (such as an optical motion-capture system or millimeter-accurate laser measurement setup) would enable direct evaluation of absolute state estimation accuracy. This complements the Normalized Innovation Squared (NIS) metric utilized in this work, which strictly assesses internal filter consistency rather than true absolute pose error.

- **Comparative Non-Linear Filtering Benchmark:** Executing a quantitative benchmark comparing the Extended Kalman Filter (EKF) against the Unscented Kalman Filter (UKF) or Particle Filter (PF) on the captured empirical datasets remains a compelling research path. This is particularly relevant when investigating filter performance during high-uncertainty regimes where first-order Taylor series linearization in the EKF introduces non-negligible truncation errors. However, given its substantial scope, this comparative study is better suited for a dedicated extended project.

Problem Scope Extensions

To transition the current prototype into a fully autonomous navigation stack, several long-term research directions are proposed:

- **Global Localization and Kidnapped Robot Resolution:** The current system addresses position tracking under the assumption of a known initial state x_0 . A natural extension involves implementing global localization algorithms-such as Monte Carlo Localization (MCL) or Particle Filtering-allowing the system to initialize and recover pose without prior spatial assumptions, thereby overcoming the single-mode Gaussian distribution limits inherent to EKF frameworks.
- **Closed-Loop Motion Control Integration:** Feeding the estimated pose output into a closed-loop trajectory tracking or path-following controller represents the next operational step. The present work strictly bounds its scope to state estimation and localization performance analysis, establishing the fundamental estimation layer required for feedback control.
- **Large-Scale Operational Scalability:** Validating system performance across significantly larger operational areas beyond the current $2\text{ m} \times 2\text{ m}$ laboratory testbed-incorporating higher ArUco landmark densities and complex occlusion dynamics-is a prerequisite prior to real-world industrial deployment.

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CHAPTER 7

APPENDIX

7.1 Nomenclature and Variable Definitions

Table 7.1 summarizes all mathematical symbols, matrix dimensions, and physical descriptions across the EKF pipeline.

Table 7.1: Extended Kalman Filter Symbol Nomenclature and Matrix Dimensions

Symbol	Dimension	Description
x_t	3×1	Posterior state vector $[p_x, p_y, \theta]^\top$ at time step t
$\bar{x}_t, \bar{P}_t$	$3 \times 1, 3 \times 3$	Prior state estimate and prior error covariance matrix
x_t, P_t	$3 \times 1, 3 \times 3$	Posterior state estimate and posterior error covariance matrix
u_t	3×1	Control input vector $[v_{x,u}, v_{y,u}, \omega_{\text{eff}}]^\top$
F_t	3×3	State transition Jacobian matrix $\partial f / \partial x$ evaluated at $\hat{x}_{t-1}, u_t$
G_t	3×3	Control input Jacobian matrix $\partial f / \partial u$ evaluated at $\hat{x}_{t-1}, u_t$
Q_u	3×3	Control space noise covariance matrix $\text{diag}(\sigma_{v_x}^2, \sigma_{v_y}^2, q_\omega)$
Q_t	3×3	State space process noise covariance matrix ($G_t Q_u G_t^\top$)
z_t	3×1	Pre-transformed measurement vector $[X_r, Y_r, \theta_r]^\top$ in map frame
H	3×3	Measurement observation matrix ($\partial h / \partial x = I_3$, constant)
J_1, J_2	3×3	First-order transformation Jacobians for coordinate propagation
R_t	3×3	Global measurement noise covariance matrix ($J_2 J_1 R_{\text{local}} J_1^\top J_2^\top$)
K_t	3×3	Optimal Kalman Gain matrix